

Theory, Modeling, and Numerical Solution of Traveling-Wave Parametric Amplifiers

A Solver-Oriented Treatise: Circuit Physics, Operator Mathematics, and
Calibration

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Abstract

This book develops, from absolute first principles, everything required to build a rigorous, fast numerical solver for traveling-wave parametric amplifiers (TWPAs) and to use that solver as the engine of a measurement-calibration pipeline. It is written for two audiences simultaneously: a reader with deep microwave or quantum-engineering background who needs the mathematics made precise, and a reader who has never seen a microwave circuit but commands linear algebra and differential equations and wants to understand what the circuit words mean in mathematical terms. Every chapter carries both a circuit-facing and a mathematically-abstract description of the same objects, with explicit dictionaries linking them.

Part I derives the circuit model from scratch: node-flux Lagrangian mechanics, the nonlinear branch laws of Josephson junctions and kinetic inductors, dispersion engineering, phase matching, and the coupled-mode theory of parametric gain. Part II is the full numerical core: harmonic balance by the alternating frequency–time method, the large-signal pump solve, globalization and continuation, network condensation, the conversion matrix (including the operator-theoretic derivation that explains why a positive-only sideband basis cannot produce gain), quantum efficiency, verification, the cross-domain solver landscape, and the full operator-mathematics treatment of the one-rail gain problem: state-space split, linearization, Schur condensation of the auxiliary pump rail, the effective primary-space operator $A_{mq}^{\text{eff}} = L_m \delta_{mq} + N_{m-q}$, and its numerics from dense reference solves through preconditioned Krylov and adjoint sensitivity. Part III connects the solver to science: the inverse problem, fabrication disorder, and amortized simulation-based inference.

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Part I

Physics and Modeling

Chapter 1

Introduction and Program

1.1 What this book is for

A traveling-wave parametric amplifier is a long, weakly dissipative, nonlinear transmission line. A strong *pump* tone propagates along it and, through the nonlinearity of Josephson junctions or a kinetic inductance, modulates the line’s effective inductance in space and time. A weak *signal* tone riding the same line is amplified by drawing energy from the pump, provided the interacting tones satisfy energy and momentum conservation—the latter being the notorious *phase-matching* condition. The device is valuable because, when engineered correctly, it adds almost the minimum noise quantum mechanics permits while providing tens of decibels of gain over several gigahertz, which is exactly what is needed to read out superconducting qubits and other quantum sensors.

This book is not primarily about the device physics for its own sake. It is about building the *computational instrument* that lets us (i) predict the gain of a given design as a function of pump frequency and pump power, (ii) fit those predictions to measured gain maps to recover the true circuit parameters, (iii) extend the model to kinetic-inductance nonlinearity, and (iv) fold in fabrication variability so that calibration becomes statistical inference over a population of devices rather than a single fit. The organizing equation of the whole enterprise is the node-flux circuit equation

$$\mathbf{C} \ddot{\boldsymbol{\phi}}(t) + \mathbf{G} \dot{\boldsymbol{\phi}}(t) + \mathbf{i}_{\text{lin}}(\boldsymbol{\phi}(t)) + \mathbf{i}_{\text{nl}}(\boldsymbol{\phi}(t)) - \mathbf{i}_{\text{src}}(t) = \mathbf{0}, \quad (1.1)$$

which we will derive in Chapter 2 and then never abandon: every solver, every observable, and every calibration objective is a functional of the periodic solutions of (1.1).

1.2 The scientific program

The work decomposes into a forward problem and an inverse problem, joined by a small number of hard numerical sub-problems.

Forward problem.

Given a circuit (topology plus parameters $\boldsymbol{\theta}$) and a pump (ω_p, I_p) , compute the small-signal scattering parameters and hence the gain $G(\omega_s; \omega_p, P_p, \boldsymbol{\theta})$ at signal frequency ω_s . Sweeping (ω_p, P_p) produces a *gain map*, the central comparison object against measurement.

Inverse problem.

Given measured gain maps from one or several fabricated devices, recover the posterior dis-

tribution over θ (junction parameters, capacitances, kinetic-inductance scale, disorder statistics). This is where simulation-based inference enters.

Between them sit three numerical bottlenecks that this book treats as first-class topics rather than implementation details: (1) finding the strongly-pumped periodic steady state at all (*convergence*); (2) doing so for circuits with 10^4 elements without the cost exploding (*scaling*); and (3) extracting actual parametric gain from the pumped state (*the conversion matrix*). Experience—including the author’s—shows that effort tends to pool on the first bottleneck while the third silently produces gain maps that contain no gain. We will be explicit about all three.

1.3 Requirements the solver must meet

It is worth stating, before any mathematics, what “a good solver” must actually do, because these requirements dictate the architecture.

1. **Produce gain.** The linearized small-signal stage must represent the phase-conjugating coupling between signal and idler; otherwise it computes loss everywhere (Chapter 10).
2. **Converge in the operating regime.** The pump solve must succeed at the powers where the device actually amplifies, which is precisely where naive Newton iteration fails. This requires globalization and continuation (Chapter 8).
3. **Scale to realistic size.** Devices have thousands of cells but only a fraction are nonlinear; the solver must iterate on the nonlinear nodes alone (Chapter 9).
4. **Be honest about convergence.** Every output must carry a status: a finite number that came from a non-converged solve is not a measurement of anything (Chapter 12).
5. **Be differentiable, or cheap enough to treat as a black box.** Calibration by gradient descent or by SBI both impose this (Chapters 23–25).

Key idea

The forward solver is two solves, not one. First a *nonlinear* solve for the periodic pump state; then a *linear* solve for the small signal riding on that state. The pump solve is where convergence lives; the linear solve is where gain lives. Conflating them is the most common source of wasted effort.

1.4 How to read this book

Part I can be read by anyone with undergraduate electromagnetism and a first course in classical mechanics; it builds the model and the analytic gain theory that we will use to *validate* the numerics. Part II is the heart and assumes comfort with linear algebra and Newton’s method. Part III assumes Part II and a little probability. Readers who only want to fix a non-converging or no-gain solver should read Chapters 10, 8, and 9 first and treat the rest as reference. A reader building the calibration pipeline should add Chapters 23–25. Notation and conversion formulas are collected in the appendices.

Chapter 2

Circuit Quantization and the Node-Flux Equation

2.1 Graph, branches, and node fluxes

Model the circuit as an oriented connected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with nodes $\mathcal{V}$ and branches $\mathcal{E}$. Choose one node as ground. The fundamental dynamical variable is the *node flux*, the time integral of the node voltage measured from ground,

$$\phi_a(t) = \int_{-\infty}^t V_a(t') dt', \quad v_a(t) = \dot{\phi}_a(t). \quad (2.1)$$

Collecting the N ungrounded nodes gives $\boldsymbol{\phi} = (\phi_1, \dots, \phi_N)^\top$. For an oriented branch $e = (a \rightarrow b)$ the branch flux is the difference $\psi_e = \phi_a - \phi_b$, and stacking all branches,

$$\boldsymbol{\psi} = \mathbf{B}^\top \boldsymbol{\phi}, \quad (2.2)$$

where $\mathbf{B} \in \mathbb{R}^{N \times E}$ is the reduced incidence matrix ($B_{ae} = +1$ if e leaves a , -1 if it enters a , 0 otherwise). The node voltages are $\mathbf{v} = \dot{\boldsymbol{\phi}}$. This choice of variable is what makes inductive (including Josephson) elements local and algebraic in $\boldsymbol{\phi}$, which is exactly what a harmonic-balance solver wants.

2.2 Lagrangian and the equation of motion

Each capacitor stores energy $\frac{1}{2}C\dot{\psi}^2$ and each inductor stores energy that is a function of its branch flux. Writing the circuit Lagrangian as kinetic (capacitive) minus potential (inductive) energy,

$$\mathcal{L}(\boldsymbol{\phi}, \dot{\boldsymbol{\phi}}) = \underbrace{\frac{1}{2} \dot{\boldsymbol{\phi}}^\top \mathbf{C} \dot{\boldsymbol{\phi}}}_{\text{capacitive}} - \underbrace{\sum_{e \in \mathcal{E}_L} U_e(\psi_e)}_{\text{inductive}}, \quad \psi_e = (\mathbf{B}^\top \boldsymbol{\phi})_e, \quad (2.3)$$

where $\mathbf{C}$ is the (node) capacitance matrix assembled from branch capacitances by $\mathbf{C} = \mathbf{B}_C \text{diag}(C_e) \mathbf{B}_C^\top$. The Euler–Lagrange equations $\frac{d}{dt} \partial_{\dot{\boldsymbol{\phi}}} \mathcal{L} - \partial_{\boldsymbol{\phi}} \mathcal{L} = \mathbf{i}_{\text{src}}$ give

$$\mathbf{C} \ddot{\boldsymbol{\phi}} + \mathbf{B}_L \mathbf{i}_L(\mathbf{B}_L^\top \boldsymbol{\phi}) = \mathbf{i}_{\text{src}}(t), \quad i_{L,e}(\psi_e) = U'_e(\psi_e), \quad (2.4)$$

i.e. each inductive branch contributes a current $i_{L,e} = U'_e(\psi_e)$ that is mapped to the nodes by the local incidence $\mathbf{B}_L$. Dissipation (port resistances, dielectric and conductor loss) is not

conservative and enters through a Rayleigh dissipation function or, equivalently, by adding a conductance term $\mathbf{G}\dot{\boldsymbol{\phi}}$ where $\mathbf{G} = \mathbf{B}_R \text{diag}(1/R_e) \mathbf{B}_R^\top$. Splitting the inductive currents into linear and nonlinear parts yields the master equation (1.1),

$$\boxed{\mathbf{C} \ddot{\boldsymbol{\phi}} + \mathbf{G} \dot{\boldsymbol{\phi}} + \mathbf{K} \boldsymbol{\phi} + \mathbf{B}_J \mathbf{i}_J(\mathbf{B}_J^\top \boldsymbol{\phi}) = \mathbf{i}_{\text{src}}(t)} \quad (2.5)$$

with the linear inverse-inductance (stiffness) matrix $\mathbf{K} = \mathbf{B}_L \text{diag}(1/L_e) \mathbf{B}_L^\top$ for ordinary inductors, augmented by coupled-inductor blocks (Section 2.3), and $\mathbf{B}_J$ the incidence of the nonlinear branches only.

Remark. Equation (2.5) is a second-order DAE in general because $\mathbf{C}$ can be singular (nodes with no capacitance to ground). In practice one regularizes by adding small parasitic capacitances or by eliminating algebraic nodes; we assume $\mathbf{C} \succ 0$ after grounding and regularization so that the system is a genuine second-order ODE. The conditioning consequences of this choice are revisited in Chapter 9.

2.3 Coupled inductors and the directional coupler

A directional coupler is, electrically, two transmission-line arms with mutually coupled inductances. For a pair of coupled branches with flux vector $\boldsymbol{\psi}_c = (\psi_1, \psi_2)^\top$ the constitutive law is $\boldsymbol{\psi}_c = \mathbf{L}_c \mathbf{i}_c$ with

$$\mathbf{L}_c = \begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix}, \quad M = k \sqrt{L_1 L_2}, \quad |k| < 1, \quad (2.6)$$

so the branch currents are $\mathbf{i}_c = \mathbf{L}_c^{-1} \boldsymbol{\psi}_c$ and the nodal contribution is the linear, frequency-independent stiffness block

$$\mathbf{i}_{\text{coupler}}(\boldsymbol{\phi}) = \mathbf{B}_c \mathbf{L}_c^{-1} \mathbf{B}_c^\top \boldsymbol{\phi}. \quad (2.7)$$

The crucial physical role of the coupler is not in the small-signal stiffness but in *where it injects the pump*: the pump enters an auxiliary arm and is coupled, with a designed magnitude and phase, into the main nonlinear line at two (or more) separated locations. The relative phase between injection points, together with the propagation phase of the line, is one knob of interferometric phase matching (Chapter 4).

2.4 Ports as sources and as scattering boundaries

A port at node p with reference impedance Z_0 has two equivalent descriptions. For nodal assembly it is a Norton source: a conductance $G_{\text{port}} = Z_0^{-1} \mathbf{u}_p \mathbf{u}_p^\top$ (with $\mathbf{u}_p$ the unit selector for node p) in parallel with a current source $\mathbf{i}_{\text{src},p}(t) = \mathbf{u}_p i_{\text{in}}(t)$. A multitone drive is

$$i_{\text{in}}(t) = \sum_m I_m \cos(\omega_m t + \alpha_m) = \text{Re} \left\{ \sum_m \tilde{I}_m e^{i\omega_m t} \right\}, \quad \tilde{I}_m = I_m e^{i\alpha_m}. \quad (2.8)$$

For scattering analysis the same port is a power-wave boundary. With port voltage V_p and inward current I_p ,

$$a_p = \frac{V_p + Z_0 I_p}{2\sqrt{Z_0}}, \quad b_p = \frac{V_p - Z_0 I_p}{2\sqrt{Z_0}}, \quad (2.9)$$

and a matched port driven by incident wave a_p is the Norton source $\mathbf{i}_{\text{src},p} = 2a_p/\sqrt{Z_0}$ in parallel with Z_0^{-1} . This equivalence (2.9) is the bridge we use repeatedly: the solver works in the nodal representation, but the deliverable— S -parameters and gain—lives in the wave representation. Appendix B collects the algebra.

Key idea

Everything the solver needs from the circuit is four assembled objects: the matrices $\mathbf{C}$, $\mathbf{G}$, $\mathbf{K}$, and the nonlinear map $\phi \mapsto \mathbf{B}_J \mathbf{i}_J(\mathbf{B}_J^\top \phi)$, plus port selectors. Block-by-block assembly of these four objects (Chapter 4) is the entire modeling interface.

Chapter 3

Nonlinear Branch Laws: Josephson and Kinetic Inductance

The single most important physical content of a TWPA is the constitutive law of its nonlinear inductive branch. Both technologies used in practice—Josephson junctions and kinetic inductors—reduce to a *current-dependent inductance*, and both can be written as a branch current that is an odd analytic function of the branch flux. Treating them under one interface is what lets the same solver serve both, which is exactly the path from a Josephson model to the kinetic-inductance extension.

3.1 The Josephson junction

A Josephson junction relates branch current to branch flux through

$$i_J(\psi) = I_c \sin\left(\frac{\psi}{\varphi_0}\right), \quad \varphi_0 = \frac{\Phi_0}{2\pi} = \frac{\hbar}{2e}, \quad (3.1)$$

where I_c is the critical current and φ_0 the reduced flux quantum. Expanding for $|\psi| \ll \varphi_0$,

$$i_J(\psi) = \frac{I_c}{\varphi_0} \psi - \frac{I_c}{6\varphi_0^3} \psi^3 + \frac{I_c}{120\varphi_0^5} \psi^5 - \dots = \frac{\psi}{L_{J0}} \left(1 - \frac{1}{6}(\psi/\varphi_0)^2 + \dots\right), \quad (3.2)$$

with the zero-bias Josephson inductance $L_{J0} = \varphi_0/I_c$. The leading nonlinearity is *cubic* and *even* in the inductance (it depends on ψ^2), so an unbiased junction is a *four-wave-mixing* (4WM, Kerr) element: the lowest process conserving photon number couples two pump photons to a signal and an idler, $\omega_s + \omega_i = 2\omega_p$. The cubic coefficient is the self/cross-phase-modulation strength that both provides gain and, through the pump's self-phase modulation, detunes the phase-matching condition—a tension we quantify in Chapter 5.

Three-wave mixing by DC bias. Apply a DC current so that $\psi = \Psi_0 + \delta\psi$ with Ψ_0 a static working point. Expanding (3.1) about Ψ_0 ,

$$i_J = I_c \sin \frac{\Psi_0}{\varphi_0} + \frac{I_c}{\varphi_0} \cos \frac{\Psi_0}{\varphi_0} \delta\psi - \frac{I_c}{2\varphi_0^2} \sin \frac{\Psi_0}{\varphi_0} \delta\psi^2 - \dots, \quad (3.3)$$

which now contains a *quadratic* term proportional to $\sin(\Psi_0/\varphi_0)$. A quadratic inductive nonlinearity is a *three-wave-mixing* (3WM) element, $\omega_s + \omega_i = \omega_p$, with the great practical advantage that signal and idler sit far from the pump, easing pump rejection. The same mechanism is engineered statically in SNAIL- and rf-SQUID-based lines, which is why a 3WM model is just (2.5) with a biased branch law.

3.2 Kinetic inductance

In a superconducting film the kinetic energy of the supercurrent gives an inductance that depends on the current itself. To leading order in a current that is small compared with a material scale current I_* ,

$$L(I) = L_0 \left[1 + \left(\frac{I}{I_*} \right)^2 + \dots \right], \quad (3.4)$$

an *even* dependence on current, hence—like the unbiased junction—a 4WM element. The branch relation $\psi = L(I) I$ can be inverted to express current as a function of flux, but it is cleaner to keep current as the natural variable and write the branch voltage $v = d\psi/dt = d[L(I)I]/dt$. For harmonic balance the convenient object is the inverse relation $i_K(\psi)$ obtained by series reversion of $\psi = L_0 I + (L_0/I_*^2)I^3$:

$$i_K(\psi) = \frac{\psi}{L_0} \left[1 - \frac{1}{I_*^2} \left(\frac{\psi}{L_0} \right)^2 + \dots \right]. \quad (3.5)$$

Comparing (3.5) with (3.2), the kinetic inductor is structurally *the same cubic branch law* as the junction, with the cubic coefficient set by I_* instead of φ_0 . A DC bias current I_{dc} again produces a quadratic term and thus 3WM, exactly as in (3.3); this is the basis of the DC-biased 3WM KI-TWPA. The differences that matter in practice are (i) I_* is much larger than I_c , so kinetic-inductance lines tolerate far higher powers and have higher dynamic range; (ii) the kinetic inductor has negligible self-capacitance compared with a junction's C_J , changing the cell's dispersion; and (iii) loss and frequency dependence of L_0 can matter at the percent level.

Key idea

Josephson and kinetic-inductance branches share one cubic, current-controlled inductive law

$$i_{nl}(\psi) = \frac{\psi}{L_0} \left[1 - \beta (\psi/\Psi_*)^2 + \dots \right], \quad (3.6)$$

with $(\beta, \Psi_*) = (\frac{1}{6}, \varphi_0)$ for a junction and a material-set pair for kinetic inductance, and a DC working point turning the leading process from 4WM into 3WM. Implementing (3.6) as the solver's nonlinear primitive means the kinetic-inductance extension is a change of two scalars and a bias, not a new solver.

Pitfall

Do not truncate (3.1) at cubic order inside the pump solve when the pump is strong. The full sin generates all odd harmonics, and a strong pump populates them; a cubic truncation both misrepresents the pump waveform and changes the effective phase matching. Use the exact sin (and the exact $L(I)$ reversion to the order you can afford) in the time-domain evaluation, and reserve the polynomial expansion for analytic intuition only.

3.3 Generalized nonlinear inductance and a unified parameter

A convenient phenomenological unification, useful when fitting unknown devices, writes the branch inductance as

$$L(I) = L_0 [1 + \beta_{nl} (I/I_*)^2], \quad (3.7)$$

with β_{nl} an $O(1)$ shape factor that interpolates between junction-like and kinetic-inductance-like nonlinearity. This is the single most useful “sanity-check” law before full harmonic balance: it lets one compute coupled-mode gain analytically (Chapter 5) and provides a low-dimensional

handle for the inverse problem (Part III), where $I_\star$ and β_{nl} are among the parameters to be inferred.

Chapter 4

From Elements to Architectures: Lines, Couplers, and Phase Matching

4.1 The Josephson transmission line as a periodic medium

A Josephson transmission line (JTL) is a ladder whose series inductors are the nonlinear branches of Chapter 3 and whose shunt elements are capacitances C_g to ground (plus the junction capacitance C_J in parallel with each series branch). For node fluxes $\phi_0, \dots, \phi_N$ with series branch fluxes $\psi_n = \phi_{n-1} - \phi_n$, the line contributes

$$\mathbf{C}_{\text{JTL}} = \mathbf{C}_g + \mathbf{D} \text{diag}(C_{J,n}) \mathbf{D}^T, \quad \mathbf{i}_{\text{JTL}}(\boldsymbol{\phi}) = \mathbf{D} \mathbf{i}_{\text{nl}}(\mathbf{D}^T \boldsymbol{\phi}), \quad (4.1)$$

where $\mathbf{D}$ is the series-branch incidence. In the small-signal, low-power limit the line is an ordinary LC ladder with per-cell inductance L_0 and capacitance C_g , characteristic impedance $Z_0 \approx \sqrt{L_0/C_g}$, and a low-pass dispersion.

4.2 Bloch dispersion and stopbands

Because the line is (designed to be) periodic, its small-signal propagation is governed by Bloch's theorem. For a uniform LC cell the per-cell ABCD matrix (series inductor $Z = i\omega L_0$, shunt $Y = i\omega C_g$) has trace condition giving the dispersion relation

$$\cos(ka) = 1 - \frac{1}{2}\omega^2 L_0 C_g, \quad (4.2)$$

with a the cell length and k the Bloch wavenumber. The line transmits where the right-hand side lies in $[-1, 1]$ and reflects (a *stopband*) where it does not. Deliberate spatial modulation of L or C with period p opens stopbands at the Bragg frequencies where forward and backward Bloch modes couple; following the synthesis viewpoint of periodic-structure and network theory, a chosen set of stopbands can be *designed* by prescribing their centers and widths and decomposing them onto a common spatial-harmonic basis of the modulation. The reason a TWPA designer cares is twofold: a stopband placed just above the pump steepens the local dispersion (a phase-matching tool, below), and a stopband placed at the pump's harmonic suppresses unwanted harmonic generation.

Definition 4.1 (Generalized dispersion relation). For a unit cell with frequency-dependent series impedance $Z(\omega)$ and shunt admittance $Y(\omega)$, the Bloch dispersion follows from requiring a nontrivial periodic solution of the cell transfer map, $\cosh(\gamma a) = 1 + \frac{1}{2}Z(\omega)Y(\omega)$ with $\gamma = ik$ for a propagating mode. Frequency-dependent elements (filter synthesis) and spatially modulated

frequency-independent elements (periodic loading) are two complementary routes to the same engineered $k(\omega)$.

4.3 Phase matching

Parametric gain accumulates along the line only if the interacting tones stay in phase. Writing the per-unit-length wavenumbers $k(\omega)$, the phase mismatch for 4WM is

$$\Delta k_{4\text{WM}} = k_s + k_i - 2k_p + \Delta k_{\text{nl}}, \quad \omega_s + \omega_i = 2\omega_p, \quad (4.3)$$

and for 3WM

$$\Delta k_{3\text{WM}} = k_s + k_i - k_p, \quad \omega_s + \omega_i = \omega_p, \quad (4.4)$$

where Δk_{nl} in (4.3) is the *nonlinear* correction from self- and cross-phase modulation, proportional to pump power. Exponential gain requires $|\Delta k| \lesssim$ the gain coefficient over the device length. Because the bare LC dispersion is convex, an unengineered 4WM line is phase-mismatched, and three families of fixes exist:

1. **Resonant phase matching:** a resonant inclusion (e.g. an *LC* trap) creates a sharp dispersion feature near the pump that cancels Δk_{nl} .
2. **Dispersion-engineered (stopband) phase matching:** a designed stopband just above the pump steepens $k(\omega_p)$ so the bare mismatch compensates the nonlinear one (the synthesis route above).
3. **Interferometric phase matching (IPM):** the pump is injected at two or more points with a controlled relative phase, so that the *effective* pump phase advances in step with the signal-idler beat regardless of the bare dispersion. This is the family the present device belongs to and the reason directional couplers are central rather than incidental.

4.4 The interferometric phase-matched (IPM) topology

The device is a chain

$$\text{Port}_{\text{in}} \rightarrow \text{Coupler}_1 \rightarrow \text{JTL}_1 \rightarrow \text{Coupler}_2 \rightarrow \text{JTL}_2 \rightarrow \text{Port}_{\text{out}}, \quad (4.5)$$

in which each coupler injects a pump tone from an auxiliary (pump) line into the main (signal) line. The two pump injection currents are

$$\mathbf{i}_{\text{src,pump}}(t) = \mathbf{u}_{p,1} I_{p,1} \cos(\omega_p t + \alpha_1) + \mathbf{u}_{p,2} I_{p,2} \cos(\omega_p t + \alpha_2), \quad (4.6)$$

and the relative phase $\alpha_2 - \alpha_1$, together with the coupler strength k and the propagation phase between injection points, are the interferometric phase-matching parameters. In the assembled model (2.5), couplers contribute linear stiffness blocks (2.7) and pump source terms (4.6); the JTL sections contribute the nonlinear assembly (4.1). Nothing in the global form of (2.5) changes with topology—only the block list, the local incidences, and the branch laws do. This invariance is the architectural license for a block-residual solver (Chapter 6 and Section 26.1).

Pitfall

The pump current that drives the junctions is *not* the source current at the pump port. With -14 dB couplers the per-junction pump amplitude is several times smaller than the port current. A “pump multiplier” quoted relative to I_c but computed from the *port* current can be misleading by the coupler factor; always read the realized junction-branch current from the converged pump state (Chapter 12). This single number decides whether a convergence failure is numerical or physical.

Chapter 5

Coupled-Mode Theory of Parametric Gain

Before any numerics, the coupled-mode equations (CME) give closed-form gain. They are indispensable not because they capture every effect—they assume an undepleted pump, a handful of modes, and a known dispersion—but because they are the analytic ground truth against which the full solver is validated, and because they make transparent *why the idler is non-negotiable*.

5.1 Three coupled modes

Let the slowly varying complex amplitudes of pump, signal, and idler along the line be $A_p(z)$, $A_s(z)$, $A_i(z)$. For a 4WM line with cubic nonlinearity coefficient γ (units of inverse power times inverse length), the standard CME in the undepleted-pump approximation are

$$\frac{dA_s}{dz} = i\gamma(2|A_p|^2 A_s + A_p^2 A_i^* e^{i\Delta k z}), \quad (5.1)$$

$$\frac{dA_i^*}{dz} = -i\gamma(2|A_p|^2 A_i^* + (A_p^*)^2 A_s e^{-i\Delta k z}), \quad (5.2)$$

where $\Delta k = \Delta k_{4WM}$ from (4.3). The two terms in (5.1) are self/cross-phase modulation ($2|A_p|^2 A_s$) and the parametric coupling ($A_p^2 A_i^*$). The decisive structural fact is that the signal couples to the *conjugate* of the idler. This is the phase-conjugating coupling; without an idler equation (5.2) to couple back, A_s merely accumulates phase and never grows.

5.2 Gain and the phase-matching condition

Define the total (linear plus nonlinear) phase mismatch $\kappa = \Delta k + 2\gamma|A_p|^2$ and the parametric coupling $g = \gamma|A_p|^2$. Solving the linear system (5.1)–(5.2) gives the signal power gain over a line of length ℓ ,

$$G_s = \left| \cosh(g_e \ell) + i \frac{\kappa}{2g_e} \sinh(g_e \ell) \right|^2, \quad g_e = \sqrt{g^2 - (\kappa/2)^2}. \quad (5.3)$$

Two regimes follow immediately. When phase matching is perfect, $\kappa = 0$, $g_e = g$ and the gain is exponential, $G_s = \cosh^2(g\ell) \sim \frac{1}{4}e^{2g\ell}$. When $|\kappa| > 2g$, g_e is imaginary, the hyperbolic functions become trigonometric, and the “gain” oscillates around unity—there is no amplification. The whole art of TWPA design, and the reason phase matching dominates Chapter 4, is to hold $\kappa \approx 0$ across the band by balancing the power-dependent $2\gamma|A_p|^2$ against an engineered Δk .

Key idea

Gain in (5.3) comes entirely from the signal–idler–conjugate coupling. A numerical small-signal scheme that does not represent the idler at ω_i (4WM: $2\omega_p - \omega_s$; 3WM: $\omega_p - \omega_s$) *and its conjugate relationship to the signal* cannot produce $G_s > 1$, no matter how well the pump is converged. This is the theoretical statement of the conversion-matrix requirement made precise in Chapter 10.

5.3 3WM, and why the idler placement matters

For 3WM the coupling is $\propto A_p A_i^*$ (one pump photon), the matched coupling is $g = \gamma_3 |A_p|$, and the idler sits at $\omega_i = \omega_p - \omega_s$. Because signal and idler are on opposite sides of $\omega_p/2$ and far apart for $\omega_s \ll \omega_p$, the device is easier to operate (pump is out of band) but the gain is linear in pump *amplitude* rather than power, and the static bias that creates the 3WM term also shifts the dispersion. For the solver these are parameter changes; for the experimentalist they are the difference between a clean and a messy gain map. We will validate the full solver against both (5.3) and its 3WM analogue in Chapter 12.

Part II

The Forward Solver

Chapter 6

Periodic Steady State and Harmonic Balance

6.1 The right problem to solve

A TWPA driven by a periodic pump settles, after transients, into a periodic steady state (PSS) at the pump period $T = 2\pi/\omega_p$. We do not care about the transient; we care about the PSS and the small-signal response around it. Time-domain integration of (2.5) until transients decay is possible but wasteful: the line is high- Q and weakly damped, so transients last many pump periods, and we would then have to Fourier-analyze the result anyway. Harmonic balance solves directly for the Fourier coefficients of the PSS, turning a stiff ODE-in-time into an algebraic system in frequency. This is the standard and correct choice for microwave periodic-steady-state problems, and it is what the established Josephson-circuit solvers do.

Definition 6.1 (Pump PSS problem). Find $\phi(t) = \phi(t + T)$ satisfying $C\ddot{\phi} + G\dot{\phi} + K\phi + B_J \mathbf{i}_J(B_J^\top \phi) = \mathbf{i}_{\text{src,pump}}(t)$, with $\mathbf{i}_{\text{src,pump}}$ the pump drive (4.6).

6.2 The harmonic-balance ansatz

Expand the periodic node fluxes in a real Fourier series truncated at H pump harmonics,

$$\phi(t) = \sum_{k=1}^H [c_k \cos(k\omega_p t) + s_k \sin(k\omega_p t)] \quad (+ c_0 \text{ if DC is present}), \quad (6.1)$$

collecting the unknowns into $\mathbf{x} = (c_1, s_1, \dots, c_H, s_H) \in \mathbb{R}^{2NH}$. Time derivatives are algebraic in this basis: $\cdot \mapsto$ multiply harmonic k by $k\omega_p$ and swap $\cos \leftrightarrow \sin$ with a sign; $\ddot{\cdot} \mapsto$ multiply by $-(k\omega_p)^2$. The linear part of (2.5) is therefore block-diagonal across harmonics, with harmonic- k block

$$A_k = K - (k\omega_p)^2 C, \quad B_k = (k\omega_p) G, \quad (6.2)$$

acting on (c_k, s_k) as $\begin{bmatrix} A_k & B_k \\ -B_k & A_k \end{bmatrix}$. The difficulty is entirely in the nonlinear term $\mathbf{i}_J$, whose Fourier coefficients are not algebraic functions of $\mathbf{x}$.

6.3 Alternating frequency–time (AFT) evaluation

The nonlinearity is local and cheap in *time* but tangled in *frequency*. The AFT method exploits this: synthesize $\phi(t)$ at N_t sample times from the coefficients, evaluate the exact branch law

pointwise, and project back onto the harmonic basis by discrete Fourier transform. Concretely, the harmonic-balance residual is

$$\mathbf{R}(\mathbf{x}) = \underbrace{\mathcal{F}[\mathcal{L}\{\phi\}]}_{\text{linear, algebraic}} + \underbrace{\mathcal{F}[\mathbf{B}_J \mathbf{i}_J(\mathbf{B}_J^\top \phi(t))]}_{\text{AFT}} - \underbrace{\mathcal{F}[\mathbf{i}_{\text{src,pump}}(t)]}_{\text{known}}, \quad (6.3)$$

where $\mathcal{F}$ denotes projection onto the truncated cosine/sine basis,

$$\mathbf{R}_k^{\cos} = \frac{2}{N_t} \sum_{m=0}^{N_t-1} \mathbf{r}(t_m) \cos(k\omega_p t_m), \quad \mathbf{R}_k^{\sin} = \frac{2}{N_t} \sum_{m=0}^{N_t-1} \mathbf{r}(t_m) \sin(k\omega_p t_m), \quad (6.4)$$

with $t_m = mT/N_t$ and $\mathbf{r}(t)$ the time-domain residual. To avoid aliasing of the harmonics that the nonlinearity generates, choose $N_t \geq 2H + 1$ and in practice oversample, $N_t \gtrsim 4(2H + 1)$. The cost of one residual evaluation is N_t pointwise branch evaluations plus N_t sparse matrix-vector products—linear in the number of elements, which is what makes the method scale.

Recipe

Residual evaluation, once: (1) synthesize $\phi(t_m), \dot{\phi}(t_m), \ddot{\phi}(t_m)$ from $\mathbf{x}$; (2) form $\mathbf{r}(t_m) = \mathbf{C}\ddot{\phi} + \mathbf{G}\dot{\phi} + \mathbf{K}\phi + \mathbf{B}_J \mathbf{i}_J(\mathbf{B}_J^\top \phi) - \mathbf{i}_{\text{src}}$; (3) project with (6.4). Scale the residual by the drive current so the convergence tolerance is dimensionless and power-independent.

6.4 The analytic Jacobian

Newton’s method needs $\partial \mathbf{R} / \partial \mathbf{x}$. The linear part contributes the frequency blocks (6.2) directly. The nonlinear part has a Jacobian that is again assembled by AFT: differentiating (6.3), the time-domain tangent of the Josephson current is the *instantaneous* stiffness

$$\mathbf{K}_J(t) = \mathbf{B}_J \text{diag}\left(\frac{I_{c,e}}{\varphi_0} \cos \frac{\psi_e(t)}{\varphi_0}\right) \mathbf{B}_J^\top, \quad \psi(t) = \mathbf{B}_J^\top \phi(t), \quad (6.5)$$

a periodic, symmetric matrix. Its contribution to the harmonic- k , harmonic- ℓ block of the Jacobian is the Fourier coefficient $\widehat{\mathbf{K}}_{J(k-\ell)}$ of (6.5), which couples harmonics differing by the harmonic order of the pump modulation. Assembling these blocks by FFT of $\mathbf{K}_J(t)$ gives the exact sparse Jacobian at the cost of a handful of residual evaluations. Using the analytic Jacobian rather than a finite-difference one is essential at scale: the finite-difference Jacobian costs $2NH$ residual evaluations and destroys sparsity.

Pitfall

A frequent and expensive mistake is to build the Jacobian by repeated sparse sub-block assignment `J[rows, cols] += ...` on a compressed-sparse-column matrix. CSC sub-assignment is near-quadratic; instead accumulate the four (or $2H \times 2H$) dense-in-harmonic, sparse-in-node blocks separately and concatenate once. The structure is block-sparse in nodes and small-dense in harmonics; respect it.

Chapter 7

The Large-Signal Pump Solve

7.1 Newton’s method and its discontents

With residual $\mathbf{R}$ and Jacobian $\mathbf{J}$, the Newton iteration is $\mathbf{x}^{(n+1)} = \mathbf{x}^{(n)} - \mathbf{J}^{-1}\mathbf{R}$. Near a solution it converges quadratically; far from one it diverges, oscillates, or steps into a region where $\mathbf{C}\ddot{\phi}$ overflows. For a weakly pumped line, $\mathbf{x} = \mathbf{0}$ is an excellent initial guess and plain Newton converges in a few steps. The trouble is exactly the regime we care about: as pump power rises toward the amplifying region, the cosine in (6.5) swings over a large range, the instantaneous stiffness can go negative on part of the cycle, the Jacobian becomes ill-conditioned or indefinite, and the basin of attraction of Newton shrinks below the size of the step needed to reach the target from $\mathbf{0}$.

7.2 Damping and the Levenberg–Marquardt safeguard

The first and cheapest fix is a damped (line-search) Newton step. Solve the regularized system

$$(\mathbf{J} + \lambda \mathbf{I}) \Delta \mathbf{x} = -\mathbf{R}, \quad (7.1)$$

and accept $\mathbf{x} + \Delta \mathbf{x}$ only if the residual norm strictly decreases; otherwise increase λ (toward gradient descent on $\frac{1}{2}\|\mathbf{R}\|^2$) and retry. Solving the square system (7.1) rather than the normal equations $(\mathbf{J}^T \mathbf{J} + \lambda \mathbf{I}) \Delta \mathbf{x} = -\mathbf{J}^T \mathbf{R}$ avoids squaring the condition number, which matters here because $\mathbf{J}$ is already ill-conditioned. A monotone residual-decrease acceptance test keeps the reported status honest: a step that does not reduce the residual is not progress.

7.3 Higher-order correctors

Newton uses only first derivatives. When the Jacobian is expensive to factorize and the bottleneck is the number of (factorization-bearing) iterations, higher-order methods that use a directional second derivative can reduce iteration count. Halley’s method, the second-order member of the Householder family, augments the Newton step with a curvature correction computable by a single additional directional derivative (efficiently obtained by Taylor-mode automatic differentiation), achieving cubic local convergence. On large, ill-conditioned nonlinear systems—of which the strongly-pumped TWPA is a textbook example—a Halley corrector can outperform Newton by spending fewer outer iterations and hence fewer Jacobian factorizations, at the cost of one extra directional second-derivative evaluation per step. This is worth considering once the

cheaper globalization of Chapter 8 is in place; it is a corrector improvement, not a substitute for continuation.

7.4 Matrix-free and Krylov solves

For very large lines the Jacobian factorization dominates. A matrix-free Newton–Krylov scheme avoids forming J explicitly: GMRES needs only Jacobian–vector products, and the AFT structure gives those cheaply because $J\mathbf{v}$ is a directional derivative of the residual, computable by one extra AFT pass (or by a finite-difference of the residual along $\mathbf{v}$). Preconditioning is then the whole game; the natural preconditioner is the harmonic-block-diagonal linear operator (6.2), optionally with the pump-averaged stiffness added, which captures the dominant structure and is cheap to factor block by block.

Key idea

Rank the levers for the pump solve by impact: (1) a good initial guess from continuation (Chapter 8); (2) reducing the system to nonlinear nodes (Chapter 9); (3) the choice of corrector (damped Newton, Halley) and of linear solver (sparse LU, Krylov). The literature and hard experience agree that (1) and (2) dominate; the inner linear-algebra choice (3), where it is tempting to spend time benchmarking, is the smallest lever.

Chapter 8

Globalization and Continuation

This chapter is the answer to “the solver does not converge in the range we need.” It almost never means the solution does not exist; it means the initial guess is outside Newton’s basin. Continuation (homotopy) constructs a sequence of easy-to-hard problems and walks the solution along it.

8.1 Natural-parameter continuation in pump power

Introduce a homotopy parameter $\mu \in [0, 1]$ scaling the pump drive, $\mathbf{i}_{\text{src,pump}} \rightarrow \mu \mathbf{i}_{\text{src,pump}}$. At $\mu = 0$ the problem is linear and the solution is $\mathbf{x} = \mathbf{0}$ exactly. Increase μ in steps $\mu_0 = 0 < \mu_1 < \dots < \mu_M = 1$, and at each step solve the pump PSS with the *previous* converged state as initial guess (*warm start*). Each step is a small perturbation, so Newton stays in-basin and converges in a few iterations. The step size is adapted: shrink $\Delta\mu$ when the corrector struggles, grow it when it converges easily. This single idea is usually the difference between a solver that fails at the operating point and one that sails through it.

Recipe

Pump-power continuation: set $\mu = 0$, $\mathbf{x} = \mathbf{0}$. Loop: predict $\mathbf{x}_{\text{guess}}$ from previous states; attempt the corrector at $\mu + \Delta\mu$; on success, accept, grow $\Delta\mu$; on failure, halve $\Delta\mu$ and retry; stop at $\mu = 1$. Cache every converged $(\mu, \mathbf{x})$ so neighboring grid cells (in a map) can warm-start each other.

8.2 Predictor–corrector and the role of Newton

A better predictor than “reuse the last state” is to extrapolate. A first-order (secant/tangent) predictor uses the rate of change of $\mathbf{x}$ with μ ,

$$\mathbf{x}_{\text{pred}}(\mu + \Delta\mu) = \mathbf{x}(\mu) + \Delta\mu \frac{d\mathbf{x}}{d\mu}, \quad \mathbf{J} \frac{d\mathbf{x}}{d\mu} = -\frac{\partial \mathbf{R}}{\partial \mu}, \quad (8.1)$$

i.e. one extra linear solve with the already-factored Jacobian gives the tangent. The corrector (damped Newton) then only cleans up the predictor’s error. In this picture Newton is demoted from “the method” to “the corrector,” which is the correct mental model: the method is the homotopy, Newton merely projects back onto the solution branch.

8.3 Arc-length continuation past turning points

Natural-parameter continuation fails where the solution branch folds back (a turning point, e.g. at the onset of a bifurcation or a parametric instability), because there $d\mathbf{x}/d\mu$ blows up. Pseudo-arclength continuation parameterizes the branch by arc length s instead of μ , appending the scalar constraint

$$\dot{\mathbf{x}}^\top(\mathbf{x} - \mathbf{x}_0) + \dot{\mu}(\mu - \mu_0) - \Delta s = 0 \quad (8.2)$$

to the residual system and solving for $(\mathbf{x}, \mu)$ jointly. This traverses folds smoothly and is the right tool if the gain map shows abrupt onset edges that natural continuation cannot cross. Whether you need it is diagnostic: if pump-power continuation stalls reproducibly at a particular power independent of step size, suspect a turning point and switch to arc length.

8.4 Pseudo-transient continuation

A robust alternative that needs no homotopy parameter is pseudo-transient continuation (a damped, implicitly time-stepped relaxation toward the steady state):

$$\left(\frac{1}{\tau_n} \mathbf{I} + \mathbf{J}(\mathbf{x}^{(n)})\right) \Delta \mathbf{x} = -\mathbf{R}(\mathbf{x}^{(n)}), \quad (8.3)$$

with a pseudo-time step τ_n grown from small (robust, gradient-like) to large (fast, Newton-like) as the residual falls. It is forgiving of bad initial guesses and a good fallback when a single hard cell resists continuation.

Key idea

Convergence is a property of the *path*, not the point. Choose a path from a trivially solvable problem to the target, take small enough steps, warm-start each, and demote Newton to a corrector. If a path stalls, change the path (arc length, pseudo-transient), not the linear solver.

Chapter 9

Network Condensation and Scaling

A realistic IPM device has of order 10^4 elements, but only a minority—the junctions—are non-linear. Iterating Newton over all nodes wastes effort on a linear subnetwork that could be solved once. Condensation removes the linear nodes from the nonlinear iteration, shrinking the Newton system by the ratio of total nodes to junction nodes (often 5–10 $\times$) and simultaneously improving conditioning.

9.1 Partition into linear and nonlinear nodes

Order the nodes so that nonlinear (junction-incident) nodes come first, $\phi = (\phi_n, \phi_\ell)$. At each harmonic k the linear operator partitions as

$$\mathbf{Y}_k = \begin{bmatrix} \mathbf{Y}_{nn} & \mathbf{Y}_{n\ell} \\ \mathbf{Y}_{\ell n} & \mathbf{Y}_{\ell\ell} \end{bmatrix}, \quad \mathbf{Y}_k = \mathbf{A}_k + i(\text{damping block}), \quad (9.1)$$

where only ϕ_n enters the nonlinear branch currents. The linear nodes satisfy $\mathbf{Y}_{\ell n}\phi_n + \mathbf{Y}_{\ell\ell}\phi_\ell = \mathbf{i}_\ell$, which we can solve for ϕ_ℓ and substitute.

9.2 The Schur complement (port/boundary reduction)

Eliminating ϕ_ℓ gives a condensed harmonic admittance seen at the nonlinear nodes,

$$\tilde{\mathbf{Y}}_k = \mathbf{Y}_{nn} - \mathbf{Y}_{n\ell}\mathbf{Y}_{\ell\ell}^{-1}\mathbf{Y}_{\ell n}, \quad (9.2)$$

the Schur complement of the linear block. Because the linear network is frequency-dependent but *fixed*, the factorization of $\mathbf{Y}_{\ell\ell}$ at each harmonic is computed once and reused for every Newton iteration and every continuation step. The nonlinear iteration then runs only on ϕ_n with the reduced operator $\tilde{\mathbf{Y}}_k$ plus the (unchanged) junction nonlinearity. This is the multiport view that classical nonlinear-microwave harmonic balance has always used: condense the linear subnetwork to its behavior at the nonlinear device terminals and iterate only there.

Recipe

Condensation, once per pump frequency: (1) assemble $\mathbf{Y}_k$ for each harmonic; (2) factor $\mathbf{Y}_{\ell\ell}$ at each k and store the factors; (3) form (or apply matrix-free) the Schur complement $\tilde{\mathbf{Y}}_k$; (4) run the pump solve over junction nodes only; (5) back-substitute $\phi_\ell = \mathbf{Y}_{\ell\ell}^{-1}(\mathbf{i}_\ell - \mathbf{Y}_{\ell n}\phi_n)$ when full node fluxes are needed. Steps (2)–(3) are amortized over all continuation steps and all Newton iterations.

Key idea

The same Schur reduction that makes the pump solve cheap also makes it better conditioned, because the eliminated linear nodes carry the benign part of the spectrum. Network condensation is the rare lever that helps convergence and speed at once; it should be the backbone of a production solver, not an afterthought.

Chapter 10

The Conversion Matrix: Small-Signal Response Around the Pump

This is the chapter that turns a converged pump state into gain. It is also where the most damaging silent bug lives: a small-signal scheme that omits the idler–conjugate coupling produces a fully finite, fully converged, completely gainless map. We build the conversion matrix so that gain is present by construction.

10.1 Linearization about the pumped state

Let $\phi_p(t)$ be the converged T -periodic pump state. Inject a weak signal and write $\phi(t) = \phi_p(t) + \delta\phi(t)$. Linearizing (2.5),

$$\mathbf{C} \delta \ddot{\phi} + \mathbf{G} \delta \dot{\phi} + \mathbf{K}(t) \delta \phi = \delta \mathbf{i}_{\text{src}}(t), \quad \mathbf{K}(t) = \mathbf{K} + \mathbf{K}_J(t), \quad (10.1)$$

with $\mathbf{K}_J(t)$ the instantaneous Josephson stiffness (6.5) evaluated on the pump state—a T -periodic matrix. Equation (10.1) is a linear *time-periodic* (Floquet) system. This is the “small signal on a periodically time-varying linear circuit, in the undepleted-pump limit” that established solvers compute, and it is exactly where parametric gain arises.

10.2 Floquet sidebands and the conjugate idler

A signal at ω_s in a pump-periodic system generates response at the Floquet comb $\omega_s + m\omega_p$, $m \in \mathbb{Z}$. Crucially, because the physical fields are real, the basis must be closed under conjugation: the negative-frequency members of the comb are the conjugates of positive ones. Expand

$$\delta\phi(t) = \sum_{m=-M}^M \delta\Phi_m e^{i(\omega_s + m\omega_p)t}, \quad \delta\Phi_{-m} \text{ tied to } \delta\Phi_m^* \text{ by reality.} \quad (10.2)$$

The pump-modulated stiffness has Fourier components $\hat{\mathbf{K}}_\ell$ (the FFT of $\mathbf{K}(t)$), and these couple sideband m to sideband $m - \ell$. Substituting (10.2) into (10.1) and matching each comb line gives the block linear system

$$\sum_{q=-M}^M \left[\mathbf{D}_m \delta_{mq} + \hat{\mathbf{K}}_{m-q} \right] \delta\Phi_q = \delta\mathbf{I}_m, \quad \mathbf{D}_m = \mathbf{K} - (\omega_s + m\omega_p)^2 \mathbf{C} + i(\omega_s + m\omega_p) \mathbf{G}, \quad (10.3)$$

which is the *conversion matrix* acting on the stacked sideband vector. The diagonal blocks D_m are the ordinary linear circuit admittance at each comb frequency; the off-diagonal blocks $\hat{K}_{m-q}$ are the parametric coupling created by the pump.

Key idea

Where the gain comes from. For 4WM the idler lives at $\omega_i = 2\omega_p - \omega_s$, which is the comb line $-\omega_s + 2\omega_p$, i.e. the $m = 2$ member of the *conjugate* branch. If the basis is a single positive comb $\{\omega_s + m\omega_p\}$ truncated at $M = 1$ and the negative-frequency idler is not represented, the parametric loop never closes and $|S_{21}| \leq 1$ everywhere. The fix is twofold: (i) include the negative-frequency sidebands and enforce the reality relation in (10.2), and (ii) take $M \geq 2$ so the 4WM idler is inside the truncation (for 3WM, the idler at $\omega_p - \omega_s$ requires $M \geq 1$ with the conjugate branch present).

Pitfall

A conversion scheme built as a single positive-frequency comb with $M = 1$, injecting only at ω_s and reading only at ω_s , computes frequency conversion and propagation loss faithfully and *cannot* produce $G > 1$. A map made this way will show a few dB of loss across the whole plane regardless of pump power. The symptom is unambiguous: “best gain” negative everywhere, even when the pump solve reports converged at all cells. The cause is the missing conjugate idler in (10.2)–(10.3), not the pump solver.

10.3 Scattering parameters and gain

Solve (10.3) once per signal frequency, exciting one incident wave at a time. For an incident signal wave $a_{q,n} = 1$ at port q , sideband n (all others zero), read the outgoing waves $b_{p,m}$ at every port p and sideband m using the power-wave relation (2.9) applied per sideband (the port voltage at comb line m is $V_{p,m} = i(\omega_s + m\omega_p)\delta\Phi_m$ at the port node). The full multi-frequency scattering matrix is

$$S_{p,m;q,n} = b_{p,m} \big|_{a_{q,n}=1, \text{ else } 0}. \quad (10.4)$$

The deliverables are particular blocks of (10.4): the *signal gain* $G_s = |S_{\text{out},0;\text{in},0}|^2$, the *idler conversion* $G_i = |S_{\text{out},m_i;\text{in},0}|^2$ at the idler sideband m_i , and the reverse and pump leakage terms. A converged pump state plus a correctly-constructed conversion matrix yields a gain map with real gain in the phase-matched band, reproducing the coupled-mode prediction (5.3) of Chapter 5 in the appropriate limit.

Recipe

Gain at one cell: (1) take the converged pump $\mathbf{x}$; (2) FFT the instantaneous stiffness $K(t)$ to get $\hat{K}_\ell$ for $|\ell| \leq 2M$; (3) for each ω_s , assemble the conversion matrix (10.3) over $m \in [-M, M]$ with $M \geq 2$, enforcing reality; (4) solve for each unit incident wave; (5) extract S via (10.4) and report G_s, G_i and the convergence status of the underlying pump solve.

Chapter 11

Quantum-Correct Outputs: Efficiency and Added Noise

The classical conversion matrix gives gain; a quantum-limited amplifier is also judged by the noise it adds. For a thesis whose endpoint is near-quantum-limited devices, the solver should be able to report quantum efficiency and added noise, not just $|S_{21}|^2$.

11.1 From power waves to photon flux

The classical scattering matrix (10.4) mixes modes at different frequencies. To speak of photons, renormalize each mode so that its amplitude carries photon flux rather than power: define $\hat{a}_m$ with $\langle \hat{a}_m^\dagger \hat{a}_m \rangle$ the photon rate, related to the power-wave amplitude by $a_m = \sqrt{\hbar\omega_m} \hat{a}_m$. In this basis the conversion matrix becomes a Bogoliubov transformation relating input and output mode operators, and the constraint that it preserve canonical commutators encodes unitarity (for a lossless, ideal device) or its violation (for loss and impedance mismatch).

11.2 Quantum efficiency and the standard quantum limit

A phase-insensitive linear amplifier must add at least half a quantum of noise referred to the input; this is the standard quantum limit, a consequence of the commutation relations and the unavoidable coupling to an idler (vacuum) mode. The quantum efficiency η measures how close a real device comes: $\eta = 1$ at the quantum limit, $\eta < 1$ for excess noise from loss, out-of-band coupling, or impedance mismatch. From the photon-flux conversion matrix one computes η as the ratio of the ideal added noise to the realized added noise, summed over the modes that the signal couples to. The practical message for the solver is that the same conversion matrix (10.3), once renormalized to photon flux, yields η directly—loss and mismatch enter through the non-unitarity of the transformation, so a solver that already includes port resistances and dielectric loss can report quantum efficiency without new physics.

Remark. The mapping from a classical harmonic-balance solution to a photon-basis (ladder-operator) description, and the resulting closed-form quantum-efficiency expressions for multi-port multi-frequency parametric circuits, are exactly the content of the X-parameter / quantum-efficiency program for JTWPA's; the present conversion matrix is the open-source-friendly route to the same quantities.

Chapter 12

Verification and Validation

A solver is only as useful as the trust you can place in its numbers. This chapter is the discipline that earns that trust: a ladder of checks from trivial limits to cross-tool comparison, and a status taxonomy that forbids passing off non-converged numbers as measurements.

12.1 The validation ladder

1. **Linear passive limit.** With the nonlinearity off, the harmonic-balance S -parameters must equal the ABCD/cascade S -parameters of the linear ladder (Appendix B).
2. **Zero pump.** With pump power $\rightarrow 0$, the conversion matrix (10.3) must become block diagonal and the signal S_{21} must equal the ordinary linear S_{21} ; idler conversion must vanish.
3. **Weak pump.** At small pump power the signal gain must approach the linear value plus a small parametric correction, and idler conversion must grow from zero—a sign and slope check against (5.3).
4. **Coupled-mode comparison.** On a uniform, well-phase-matched line the full solver must reproduce the closed-form gain (5.3) as a function of pump power and detuning.
5. **Jacobian check.** On a small fixture, the analytic Jacobian of Section 6.4 must match a finite-difference Jacobian to several digits.
6. **Cross-tool comparison.** Where an established solver converges (lower power, modest size), the pump state and the gain curve must agree with it. Agreement *there* is the licence to trust the new solver *beyond* there, in the regime the established tool cannot reach.

Key idea

The scientific argument for a custom solver is precisely step 6 inverted: you validate against the trusted tool in the regime where both run, then claim the custom solver in the regime where only it converges. Without the agreement region, a result in the hard regime is unfalsifiable. Establishing that overlap is therefore the highest-value experiment in the whole solver effort.

12.2 A convergence status taxonomy

Every produced number must carry a status, and only one status is usable as a measurement.

VALID_CONVERGED

residual below tolerance; S finite; usable as data and as a fitting target.

FINITE_NONCONVERGED

S finite but the pump solve did not reach tolerance; usable for debugging and visualization only, never as a fit target.

NONFINITE

NaN/Inf in S or gain; unusable.

FAIL

exception, singular system, missing output; unusable.

A gain map must be presentable in three layers: the raw finite map, the convergence-masked map, and the non-converged overlay. A map most of whose cells are finite-but-non-converged is not a reproduction of anything, and presenting it as one is the most common way to mislead oneself.

Pitfall

Finite is not converged. A nonlinear solve that exits on maximum iterations can still return a finite, plausible-looking S -matrix. If that value seeds a fit or an optimization, the fit is fitting noise. Gate every objective evaluation on **VALID_CONVERGED**.

Chapter 13

The One-Rail Gain Problem: Physics, State Space, and Two-Stage Structure

13.1 Motivation and scope

The preceding chapters assembled a general solver: harmonic balance, Newton continuation, Schur condensation, and the conversion matrix. This chapter and the three that follow zoom in on one specific, important topology—the one-rail pump-coupler device—and develop its mathematical structure to a level of precision that guides every numerical decision. The goal is a treatment that is useful to two readers simultaneously.

The *circuit-engineering reader* wants to know: given the physical device (nonlinear transmission line on the signal rail, a separate linear pump-feed network that injects pump current at two coupler positions), what is the rigorous formula for the small-signal power gain, and exactly which matrix problem must be solved to evaluate it?

The *mathematical reader* wants to know: what are the spaces, the operators, and the structure? Is the operator self-adjoint? Toeplitz? Block-diagonal? What does the gain formula look like after all circuit words are removed, and what is the correct numerical strategy for the resulting structured complex linear system?

Both questions are answered by the same derivation, and we carry both viewpoints simultaneously so neither reader has to translate.

13.2 What the one-rail topology is, from scratch

If you have never seen a microwave circuit, here is what the device is. Imagine a long chain of tiny nonlinear inductors connected in series, each shunted to ground through a small capacitor. This is the signal rail, also called the main nonlinear transmission line. A second, entirely linear chain—the pump rail—runs alongside it. At two positions along the device, the two chains are weakly coupled by mutual inductance (a directional coupler). A strong oscillating current (the pump) is injected into the pump rail at one or both of these coupling positions, propagates along the pump rail to the second coupler, and is partially transferred onto the signal rail at both points. The relative phase of the pump at the two injection points is the primary design knob for controlling gain.

The signal—a much weaker oscillating current at a different frequency—enters the signal rail at one end, is amplified by exchanging energy with the pump through the nonlinear inductors, and

exits the other end. The amplification process is parametric: the pump modulates the effective inductance of the nonlinear chain in time, and this time-varying inductance couples the signal to an idler frequency, creating a closed energy-exchange loop that produces exponential growth.

Key idea

The device is physically two-rail (signal rail and pump rail), but because the pump rail is linear, it can be eliminated algebraically, leaving an effectively one-rail problem. The pump rail's only surviving influence is a frequency-dependent linear loading correction on the signal rail, plus a prescribed pump-amplitude profile at the coupler positions. This elimination is exact in the linear approximation and is the first major simplification.

13.3 The quantity called gain

The gain is not a fixed number; it depends on which pump frequency and power one chooses. For fixed operating conditions, define the small-signal power gain as

$$G_s(\omega_s; \omega_p, I_p) = |S_{P_2,0;P_1,0}|^2. \quad (13.1)$$

This is one entry of the conversion scattering matrix. P_1 is the signal input port, P_2 is the signal output port, ω_s is the signal angular frequency, ω_p is the pump angular frequency, and I_p is the pump amplitude. The subscript 0 on each port label indicates the *signal-frequency* sideband rather than an idler or harmonic. Operationally: inject a unit-amplitude signal wave at P_1 at frequency ω_s ; the pump continues to run; measure the outgoing wave at P_2 at the same signal frequency. The squared modulus of that complex ratio is the power gain.

It is critical to separate two statements that circuit engineers sometimes compress into one:

- The *gain extraction* is a **linear** operation once the pump-dependent operating state is known. No nonlinear solve is needed at the signal stage.
- The *construction of the pump-dependent state* is a **nonlinear** periodic problem—the hard part. Its output is a set of Fourier coefficients that feed directly into the linear gain stage.

This two-stage structure—**nonlinear pump solve** → **linear gain extraction**—is the central architectural fact of the whole solver and will appear repeatedly.

13.4 State variables: primary rail and auxiliary pump network

Use the node-flux formalism of Chapter 2. Collect the node fluxes of the nonlinear signal rail into the primary vector ϕ and the node fluxes of the pump-feed and coupler network into the auxiliary vector η . The full internal state is

$$\mathbf{x} = \begin{pmatrix} \phi \\ \eta \end{pmatrix}, \quad (13.2)$$

living in the direct-sum space $X = \Phi \oplus H$, where Φ is the primary space and H is the auxiliary space. This split is the *first structural fact*: it is more fundamental than any physical names.

The pump-dependent periodic operating state is

$$\mathbf{x}_p(t) = \begin{pmatrix} \phi_p(t) \\ \eta_p(t) \end{pmatrix}, \quad (13.3)$$

where the subscript p means “induced by the pump” and applies to both rails. Even though the pump is injected on the auxiliary rail, it induces a response $\phi_p(t)$ on the primary rail through the coupler. This induced primary-rail pump waveform is what modulates the nonlinear inductors and drives the parametric process.

13.5 The governing equation in full generality

Assembling the block stamps of Chapter 2, the global node-flux equation for the device is

$$\mathbf{C} \ddot{\mathbf{x}} + \mathbf{G} \dot{\mathbf{x}} + \mathbf{i}_{\text{ind}}(\mathbf{x}) = \mathbf{i}_{\text{src}}(t), \quad (13.4)$$

where the inductive current splits as $\mathbf{i}_{\text{ind}}(\mathbf{x}) = \mathbf{K}_{\text{lin}} \mathbf{x} + \mathbf{i}_{\text{nl}}(\mathbf{x})$, with $\mathbf{K}_{\text{lin}}$ containing all linear stiffness (couplers, linear inductors) and $\mathbf{i}_{\text{nl}}$ containing only the nonlinear Josephson or kinetic-inductance branch currents on the primary rail. In block form,

$$\mathbf{C} = \begin{pmatrix} \mathbf{C}_{\phi\phi} & \mathbf{C}_{\phi\eta} \\ \mathbf{C}_{\eta\phi} & \mathbf{C}_{\eta\eta} \end{pmatrix}, \quad \mathbf{G} = \begin{pmatrix} \mathbf{G}_{\phi\phi} & \mathbf{G}_{\phi\eta} \\ \mathbf{G}_{\eta\phi} & \mathbf{G}_{\eta\eta} \end{pmatrix}, \quad \mathbf{K}_{\text{lin}} = \begin{pmatrix} \mathbf{K}_{\phi\phi} & \mathbf{K}_{\phi\eta} \\ \mathbf{K}_{\eta\phi} & \mathbf{K}_{\eta\eta} \end{pmatrix}. \quad (13.5)$$

The nonlinear term lives only in the primary block:

$$\mathbf{i}_{\text{nl}}(\mathbf{x}) = \begin{pmatrix} \mathbf{B}_\phi \hat{\mathbf{i}}(\mathbf{B}_\phi^\top \phi) \\ \mathbf{0} \end{pmatrix}, \quad (13.6)$$

where $\mathbf{B}_\phi$ is the branch-to-node incidence for the nonlinear branches and $\hat{\mathbf{i}}$ is the (possibly nonlinear) branch current law. This is the second structural fact: *the nonlinearity is confined entirely to the primary space.*

The source term is

$$\mathbf{i}_{\text{src}}(t) = \begin{pmatrix} \mathbf{i}_{\text{signal}}(t) \\ \mathbf{i}_{\text{pump}}(t) \end{pmatrix}, \quad (13.7)$$

with the pump source on the auxiliary rail and the signal source on the primary rail. Port conductances enter $\mathbf{G}$.

13.6 The differential stiffness: Josephson and kinetic-inductance cases

The bridge between the nonlinear governing equation and the linear gain problem is the *differential inverse inductance*, the Jacobian of the nonlinear branch current with respect to the branch flux, evaluated on the pump-driven operating state:

$$\Gamma(t) = \left. \frac{\partial \hat{\mathbf{i}}}{\partial \psi} \right|_{\psi = \mathbf{B}_\phi^\top \phi_p(t)}. \quad (13.8)$$

This is a time-periodic diagonal matrix (diagonal because each branch is local).

For a **Josephson** branch, $i_n = I_c \sin(\psi_n/\varphi_0)$, so

$$\Gamma_n(t) = \frac{I_c}{\varphi_0} \cos\left(\frac{\psi_{p,n}(t)}{\varphi_0}\right), \quad \psi_{p,n}(t) = (\mathbf{B}_\phi^\top \phi_p(t))_n. \quad (13.9)$$

This is the linearized inverse inductance $1/L_J$ of the junction, modulated in time because the pump sweeps the junction through different operating points on the cosine.

For a **kinetic-inductance** branch with constitutive relation $\psi_n = L_0(1 + i_n^2/I_*^2)i_n$, differentiate with respect to i_n and invert (the branch current in the pump state is $i_{p,n}$):

$$\Gamma_n(t) = \left[L_0 \left(1 + \frac{3 i_{p,n}^2(t)}{I_*^2} \right) \right]^{-1}. \quad (13.10)$$

At zero pump, $\Gamma_n = 1/L_0$, the linear inductance. As the pump grows, $i_{p,n}$ grows, and Γ_n decreases (the inductance increases), creating the time-varying modulation that drives the parametric process.

Key idea

Both branch laws produce the same mathematical object: a time-periodic scalar function $\Gamma_n(t)$ at each nonlinear branch. The physics differs (cosine for JJ, polynomial for KI), but the mathematical structure of the resulting gain problem is identical. The solver handles both by swapping only the formula for Γ_n and the formula for the pump state.

13.7 The two-stage character of the full problem, stated precisely

Stage 1 is the **pump periodic steady state**. Find $\mathbf{x}_p(t) = \mathbf{x}_p(t + T)$, $T = 2\pi/\omega_p$, satisfying (13.4) with the pump source and no signal. In Fourier coordinates $\mathbf{x}_p(t) = \sum_k \mathbf{X}_k e^{ik\omega_p t}$, this becomes a nonlinear algebraic system $\mathcal{R}(\{\mathbf{X}_k\}) = \mathbf{0}$ that is solved by Newton-under-continuation (Chapters 7–8).

Stage 2 is the **linear conversion-matrix solve**. Given $\mathbf{x}_p(t)$ from Stage 1, compute $\Gamma(t)$ from (13.8), build the time-periodic linearized stiffness

$$\mathbf{K}(t) = \mathbf{K}_{\text{lin}} + \begin{pmatrix} \mathbf{B}_\phi \Gamma(t) \mathbf{B}_\phi^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}, \quad (13.11)$$

expand the small perturbation in sidebands, build the conversion matrix, solve the linear system, and read off the gain. This is the subject of the next three chapters.

$$\boxed{\text{nonlinear pump solve} \longrightarrow \Gamma(t) \text{ from (13.8)} \longrightarrow \text{linear conversion matrix} \longrightarrow G_s(\omega_s).} \quad (13.12)$$

Chapter 14

Linearization, Sideband Expansion, and the Conversion Matrix as a Structured Operator

14.1 Linearization around the pump state

Write $\mathbf{x}(t) = \mathbf{x}_p(t) + \delta\mathbf{x}(t)$ and substitute into (13.4). The zeroth-order (pump) terms cancel by construction. The first-order (signal) terms give the linearized time-periodic equation:

$$\mathbf{C} \delta\ddot{\mathbf{x}} + \mathbf{G} \delta\dot{\mathbf{x}} + \mathbf{K}(t) \delta\mathbf{x} = \delta\mathbf{i}_{\text{src}}(t), \quad (14.1)$$

with $\mathbf{K}(t)$ as in (13.11). This is a *linear time-periodic* (Floquet) system. The coefficients are periodic at the pump period T , and the signal enters as a weak external drive. Because (14.1) is linear, it can be solved exactly once $\mathbf{K}(t)$ is known.

14.2 Why the sideband comb is the right basis

In a system with periodic coefficients at frequency ω_p , any particular-frequency input at ω_s generates responses at the full Floquet comb $\{\omega_s + m\omega_p, m \in \mathbb{Z}\}$. This is Floquet's theorem applied to a driven system: the response at any one frequency couples to the response at all other frequencies in the comb. There is no way to isolate a single frequency. The right computational basis is therefore all sideband frequencies simultaneously. Truncating to $m \in \{-M, \dots, M\}$ gives a finite-dimensional approximation.

For a **physical** (real-valued) signal, the components at $+\omega$ and $-\omega$ are conjugates of each other. This reality constraint means that the expansion

$$\delta\mathbf{x}(t) = \sum_{m=-M}^M \delta\mathbf{X}_m e^{i\Omega_m t}, \quad \Omega_m = \omega_s + m\omega_p, \quad \delta\mathbf{X}_{-m} = \delta\mathbf{X}_m^* \quad (14.2)$$

is not a free expansion: the negative- m components are determined by the positive- m ones. This is the origin of the idler: the comb member at m is the idler at $\omega_s - \omega_p$ (for 3WM) or the member at $m = 2$ at $2\omega_p - \omega_s$ (for 4WM, where the idler lives at the conjugate-branch sideband). The reality constraint is what makes the gain loop close: the signal at ω_s couples to the conjugate idler, and the conjugate idler couples back. Without both positive and negative comb members in the basis, the coupling loop cannot close and the gain cannot appear.

14.3 The conversion matrix in full

Substituting (14.2) into (14.1) and projecting onto $e^{i\Omega_m t}$ for each m gives the block-linear system:

$$\sum_{q=-M}^M \mathbf{A}_{mq} \delta \mathbf{X}_q = \delta \mathbf{I}_m, \quad (14.3)$$

where the off-diagonal (mixing) and diagonal (non-mixing) parts are

$$\mathbf{A}_{mq} = \underbrace{[-\Omega_m^2 \mathbf{C} + i\Omega_m \mathbf{G}] \delta_{mq}}_{\text{sideband-diagonal: linear response at } \Omega_m} + \underbrace{\hat{\mathbf{K}}_{m-q}}_{\text{sideband-mixing: convolution by pump harmonics}}, \quad (14.4)$$

and $\hat{\mathbf{K}}_\ell$ is the ℓ -th Fourier coefficient of $\mathbf{K}(t)$:

$$\hat{\mathbf{K}}_\ell = \frac{1}{T} \int_0^T \mathbf{K}(t) e^{-i\ell\omega_p t} dt = \mathbf{K}_{\text{lin}} \delta_{\ell 0} + \begin{pmatrix} \mathbf{B}_\phi \hat{\Gamma}_\ell \mathbf{B}_\phi^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}. \quad (14.5)$$

The zero-th coefficient $\hat{\mathbf{K}}_0$ is the DC component of the time-varying stiffness; the higher Fourier coefficients $\hat{\mathbf{K}}_\ell$ for $\ell \neq 0$ are the pump-harmonic sidebands that perform mode conversion.

Now expose the $\Phi \oplus H$ block structure explicitly:

$$\mathbf{A}_{mq} = \begin{pmatrix} \mathbf{D}_{\phi\phi}(\Omega_m) \delta_{mq} + \hat{\mathbf{K}}_{m-q}^{\text{nl}} & \mathbf{D}_{\phi\eta}(\Omega_m) \delta_{mq} \\ \mathbf{D}_{\eta\phi}(\Omega_m) \delta_{mq} & \mathbf{D}_{\eta\eta}(\Omega_m) \delta_{mq} \end{pmatrix}, \quad (14.6)$$

where the frequency-domain linear blocks are

$$\mathbf{D}_{ab}(\Omega) = -\Omega^2 \mathbf{C}_{ab} + i\Omega \mathbf{G}_{ab} + \mathbf{K}_{ab}^{\text{lin}}, \quad (14.7)$$

and $\hat{\mathbf{K}}_{m-q}^{\text{nl}}$ is the sideband-mixing part restricted to the primary subspace (it carries the nonlinear stiffness Fourier coefficients). The off-diagonal coupling blocks $\mathbf{D}_{\phi\eta}$ and $\mathbf{D}_{\eta\phi}$ are sideband-*diagonal*: the coupler is linear in the small-signal equation, so it mixes rails but not frequencies. Only $\hat{\mathbf{K}}_{m-q}^{\text{nl}}$ mixes sidebands, and it lives entirely in the $\Phi \rightarrow \Phi$ block.

Key idea

The structure of $\mathbf{A}_{mq}$ has two independent axes:

1. The **internal space axis** (Φ vs. H): the coupler provides off-diagonal linear coupling between rails, but no sideband mixing.
2. The **sideband axis** (m vs. q): sideband mixing comes only from the nonlinear stiffness $\hat{\mathbf{K}}_{m-q}^{\text{nl}}$, and only in the primary block.

These two structures are completely independent. A mathematician would say: the operator has a block-diagonal sideband structure *except* for a Toeplitz perturbation localized in the primary subspace.

14.4 Input, output, and the gain formula

A unit incident signal wave at port P_1 , sideband $m = 0$, is a Norton current source $\delta \mathbf{I}_m = 2\sqrt{Y_0} \mathbf{u}_{P_1} \delta_{m0}$, where $Y_0 = 1/Z_0$ and $\mathbf{u}_{P_1}$ is the unit selector for the P_1 node. Define the embedding $E_0 : X \rightarrow \mathcal{X}$ that places a state vector into sideband block $m = 0$. Then the right-hand side is

$$\mathbf{r} = 2\sqrt{Y_0} E_0 \mathbf{u}_{P_1}. \quad (14.8)$$

The formal solution of $\mathbf{A} \delta \mathbf{X} = \mathbf{r}$ is $\delta \mathbf{X} = \mathbf{A}^{-1} \mathbf{r}$ (understood as a linear solve). The outgoing wave at P_2 , sideband $m = 0$, extracted via the port-to-wave relation $b_{P_2,0} = \sqrt{Y_0} V_{P_2,0} = \sqrt{Y_0} i \omega_s \mathbf{u}_{P_2}^\top E_0^\top \delta \mathbf{X}$, gives

$$b_{P_2,0} = 2Y_0 i \omega_s \mathbf{u}_{P_2}^\top E_0^\top \mathbf{A}^{-1} E_0 \mathbf{u}_{P_1}. \quad (14.9)$$

Define the output functional $\boldsymbol{\ell}^\top = \mathbf{u}_{P_2}^\top E_0^\top$ and the input vector $\mathbf{r} = E_0 \mathbf{u}_{P_1}$ (absorbing the normalization for compactness). Then the signal power gain is:

$$\boxed{G_s(\omega_s) = |2Y_0 i \omega_s \boldsymbol{\ell}^\top \mathbf{A}^{-1} \mathbf{r}|^2 = 4Y_0^2 \omega_s^2 |T(\omega_s)|^2, \quad T(\omega_s) = \boldsymbol{\ell}^\top \mathbf{A}^{-1} \mathbf{r}.} \quad (14.10)$$

The **transfer amplitude** $T(\omega_s)$ is a scalar bilinear form of the inverse of a structured operator. Gain computation is reduced to computing this single scalar.

Pitfall

$\mathbf{A}^{-1}$ in equation (14.10) is *analytic* notation meaning “solve the system $\mathbf{A} \mathbf{y} = \mathbf{r}$.” Never form the explicit matrix inverse. For any system of practical size, forming the inverse is slower, less accurate, and more memory-intensive than solving the system. The correct reading is always: solve, then project.

Chapter 15

Eliminating the Pump Rail: Schur Condensation to the Effective Primary Operator

15.1 Why and when the auxiliary variables can be eliminated exactly

In the small-signal equation (14.3)–(14.6), the auxiliary block $\mathbf{D}_{\eta\eta}(\Omega_m)$ is sideband-diagonal and invertible at almost all frequencies (it represents a linear network without resonances at most operating points). This means the auxiliary variables $\delta\mathbf{H}_m$ can be solved for exactly in terms of the primary variables $\delta\mathbf{\Phi}_m$, and the result substituted back. The elimination is exact, not approximate, as long as the auxiliary network is linear in the small-signal problem—which it is by construction.

This is the same Schur condensation introduced in Chapter 9 for eliminating passive linear nodes from the Newton iteration, now applied at the level of the entire pump rail.

15.2 Deriving the Schur complement

For each sideband m , the lower block row of (14.3) (with $\delta\mathbf{I}_m^\eta = \mathbf{0}$ if the signal is injected only on the primary rail) gives:

$$\mathbf{D}_{\eta\phi}(\Omega_m) \delta\mathbf{\Phi}_m + \mathbf{D}_{\eta\eta}(\Omega_m) \delta\mathbf{H}_m = \mathbf{0}. \quad (15.1)$$

Solving:

$$\delta\mathbf{H}_m = -\mathbf{D}_{\eta\eta}^{-1}(\Omega_m) \mathbf{D}_{\eta\phi}(\Omega_m) \delta\mathbf{\Phi}_m. \quad (15.2)$$

Substituting into the upper block row:

$$\sum_{q=-M}^M \left[\mathbf{D}_{\phi\phi}(\Omega_m) \delta_{mq} + \hat{\mathbf{K}}_{m-q}^{\text{nl}} - \mathbf{D}_{\phi\eta}(\Omega_m) \mathbf{D}_{\eta\eta}^{-1}(\Omega_m) \mathbf{D}_{\eta\phi}(\Omega_m) \delta_{mq} \right] \delta\mathbf{\Phi}_q = \delta\mathbf{I}_m^\phi. \quad (15.3)$$

Define the **self-energy** (the term borrowed from quantum field theory, where eliminated degrees of freedom leave behind an induced interaction on the remaining system):

$$\Sigma(\Omega_m) = \mathbf{D}_{\phi\eta}(\Omega_m) \mathbf{D}_{\eta\eta}^{-1}(\Omega_m) \mathbf{D}_{\eta\phi}(\Omega_m). \quad (15.4)$$

Then define the sideband-diagonal effective primary-space operator:

$$\mathbf{L}_m = \mathbf{D}_{\phi\phi}(\Omega_m) - \Sigma(\Omega_m), \quad (15.5)$$

and the sideband-mixing operator (a Toeplitz convolution in sideband index):

$$\mathbf{N}_{m-q} = \hat{\mathbf{K}}_{m-q}^{\text{nl}}. \quad (15.6)$$

Then (15.3) takes the compact form:

$$\sum_{q=-M}^M \mathbf{A}_{mq}^{\text{eff}} \delta \Phi_q = \delta \mathbf{I}_m^{\phi}, \quad \boxed{\mathbf{A}_{mq}^{\text{eff}} = \mathbf{L}_m \delta_{mq} + \mathbf{N}_{m-q}}. \quad (15.7)$$

This is the skeleton of the mathematical problem after all simplifications. It is worth pausing on what each term does.

$\mathbf{L}_m \delta_{mq}$ — **the sideband-diagonal non-mixing part.**

$\mathbf{L}_m$ is a complex matrix acting in the primary space Φ at sideband m . It contains two contributions: (1) $\mathbf{D}_{\phi\phi}(\Omega_m)$, the direct linear response of the primary rail at sideband frequency Ω_m , including capacitance, conductance, and linear inductance; and (2) $-\Sigma(\Omega_m)$, the *self-energy* or effective loading induced by the eliminated pump rail. The self-energy encodes how the pump rail modifies the boundary conditions of the signal rail at each sideband frequency. Because $\mathbf{D}_{\eta\eta}(\Omega_m)$ is the frequency response of the linear pump network (a rational function of Ω_m), the self-energy is a rational matrix function, and $\mathbf{L}_m$ captures the correct frequency-dependent loading including any resonances of the pump network.

$\mathbf{N}_{m-q}$ — **the sideband-mixing Toeplitz part.**

$\mathbf{N}_{m-q} = \hat{\mathbf{K}}_{m-q}^{\text{nl}}$ depends only on the difference $m - q$, so the operator $\mathbf{v} \mapsto \sum_q \mathbf{N}_{m-q} \mathbf{v}_q$ is a *discrete convolution* in sideband index. In functional-analysis language, it is a Toeplitz operator on the finite sideband lattice. This is the operator that performs mode conversion: it takes amplitude in sideband q and spreads it to sideband m via the Fourier components of the pump-modulated stiffness. It lives entirely in the primary space Φ and is nonzero only for $m \neq q$.

Key idea

After eliminating the pump rail, the gain problem is purely in the primary space Φ , with the structure $\mathbf{A}_{mq}^{\text{eff}} = \mathbf{L}_m \delta_{mq} + \mathbf{N}_{m-q}$. This is a block operator that is: (a) block-diagonal in sideband index, augmented by (b) a Toeplitz/convolution perturbation. The first part is easy to invert; the second part is what mixes modes and creates gain. This structure is fundamental to every efficient numerical method.

15.3 Physical meaning of the self-energy

The self-energy $\Sigma(\Omega)$ deserves more attention because it is easy to miss and important in practice. It is the frequency-dependent effective admittance that the signal rail “sees” at each coupler position, looking into the pump rail. If the pump rail is a matched transmission line, $\Sigma(\Omega)$ is simply a real conductance loading the signal rail. If the pump rail has resonances (e.g., from the coupler geometry or from impedance mismatches), $\Sigma(\Omega)$ has poles and the signal rail sees an inductive or capacitive load at those frequencies. Either way, omitting the self-energy is equivalent to assuming the pump rail is invisible to the signal, which is incorrect.

Numerically, $\Sigma(\Omega)$ must be computed without forming $D_{\eta\eta}^{-1}$ explicitly. The correct sequence (from the pseudocode perspective) is:

$$\Sigma(\Omega)\mathbf{v} = D_{\phi\eta}(\Omega)\mathbf{w}, \quad \text{where } D_{\eta\eta}(\Omega)\mathbf{w} = D_{\eta\phi}(\Omega)\mathbf{v}. \quad (15.8)$$

That is: given primary input $\mathbf{v}$, propagate it to the auxiliary rail via $D_{\eta\phi}$, solve the auxiliary system, then propagate back via $D_{\phi\eta}$. One linear solve of the auxiliary system per evaluation; no matrix inverse ever formed.

15.4 The reduced effective gain formula

With the notation $\ell_\phi^\top = \mathbf{u}_{P_2}^\top E_{0,\phi}^\top$ and $\mathbf{r}_\phi = E_{0,\phi}\mathbf{u}_{P_1}$ (the primary-space components of the input/output selectors), the gain in (14.10) becomes:

$$G_s(\omega_s) = 4Y_0^2\omega_s^2 |\ell_\phi^\top (\mathbf{A}^{\text{eff}})^{-1} \mathbf{r}_\phi|^2. \quad (15.9)$$

This is the exact same formula, now computed in a space roughly half the size, with the pump rail's effect absorbed into $\mathbf{L}_m$ via $\Sigma(\Omega_m)$.

15.5 Full operator classification table

Symbol	Mathematical role	Internal action	Sideband action
Φ	Primary state space	main variables	—
H	Auxiliary state space	aux. variables	—
m, q	Sideband indices in $\mathcal{M}$	—	label blocks
$\Omega_m = \omega_s + m\omega_p$	Indexed scalar frequency	—	diagonal in m
$C_{ab}, G_{ab}, K_{ab}^{\text{lin}}$	2nd/1st/0th order coeff. matrices	$b \rightarrow a$	diagonal
$D_{ab}(\Omega)$	Matrix quadratic polynomial in Ω	$b \rightarrow a$	diagonal
$D_{\phi\eta}, D_{\eta\phi}$	Interspace coupling (coupler)	$\Phi \leftrightarrow H$	diagonal
$\hat{K}_{m-q}^{\text{nl}}$	Toeplitz/convolution sideband mixer	$\Phi_q \rightarrow \Phi_m$	mixing
A_{mq}	Full conversion matrix block	$X_q \rightarrow X_m$	diag.+mixing
$\Sigma(\Omega_m)$	Schur self-energy from aux. elimination	$\Phi \rightarrow \Phi$	diagonal
$\mathbf{L}_m = D_{\phi\phi} - \Sigma$	Effective primary diagonal block	$\Phi_m \rightarrow \Phi_m$	diagonal
$\mathbf{N}_{m-q}$	Pump-induced sideband-mixing block	$\Phi_q \rightarrow \Phi_m$	mixing
$E_0\mathbf{u}_{P_1}$	Input vector	one direction at $m = 0$	sideband 0 only
$\mathbf{u}_{P_2}^\top E_0^\top$	Output functional	one direction at $m = 0$	sideband 0 only
$T = \ell^\top (\mathbf{A}^{\text{eff}})^{-1} \mathbf{r}$	Transfer amplitude	scalar	scalar
$G_s = 4Y_0^2\omega_s^2 T ^2$	Signal power gain	scalar observable	scalar

Chapter 16

Solving the Gain System: Numerical Methods, Structure, and Diagnostics

16.1 The problem is a structured linear system, not an inverse

The gain formula (15.9) contains $(\mathbf{A}^{\text{eff}})^{-1}$. This is analytic shorthand. The numerical task is:

1. Build or apply $\mathbf{A}^{\text{eff}}(\omega_s)$.
2. Solve $\mathbf{A}^{\text{eff}}\mathbf{y} = \mathbf{r}_\phi$.
3. Compute the scalar $T = \ell_\phi^\top \mathbf{y}$.
4. Return $G_s = 4Y_0^2 \omega_s^2 |T|^2$.
5. Report the relative residual $\|\mathbf{A}^{\text{eff}}\mathbf{y} - \mathbf{r}_\phi\|/\|\mathbf{r}_\phi\|$.

Step 5 is mandatory. A finite T computed from a non-converged solve is not a gain value; it is numerical noise. The `VALID_CONVERGED` discipline of Chapter 12 applies here identically.

16.2 The natural operator split and its implication for methods

The effective operator splits as

$$\mathbf{A}^{\text{eff}} = \mathbf{L} + \mathbf{N}, \quad (16.1)$$

where $\mathbf{L}_{mq} = \mathbf{L}_m \delta_{mq}$ is block-diagonal in sideband index (easy to invert: decouple into $2M + 1$ independent primary-space solves) and $\mathbf{N}_{mq} = \mathbf{N}_{m-q}$ is the Toeplitz sideband mixer (cannot be diagonalized simply, but has convolution structure that can be exploited). This split $\mathbf{A} = \mathbf{L} + \mathbf{N}$ is the natural one for all efficient methods.

16.2.1 Method 0: Dense reference solve (tiny systems only)

Assemble the full $(2M + 1)|\Phi| \times (2M + 1)|\Phi|$ matrix explicitly and apply dense LU. This is indispensable for verifying sideband indexing, sign conventions, and normalization. It is impractical for $|\Phi| \gtrsim 100$ or $M \gtrsim 5$.

16.2.2 Method 1: Sparse direct solve (small to medium)

Assemble $\mathbf{A}^{\text{eff}}$ as a sparse matrix (it is sparse because $\mathbf{L}_m$ is sparse in node space, and $\mathbf{N}_{m-q}$ is also sparse there). Apply sparse LU with fill-reducing permutation ($PAQ = LU$). This is the robust baseline for up to a few thousand unknowns and is the recommended starting point before any iterative method.

16.2.3 Method 2: Matrix-free Krylov (large systems)

For large problems, do not assemble $\mathbf{A}^{\text{eff}}$ explicitly. GMRES (the standard choice for complex, nonsymmetric, indefinite systems) needs only matrix-vector products $\mathbf{v} \mapsto \mathbf{A}^{\text{eff}}\mathbf{v}$. From (15.7):

$$(\mathbf{A}^{\text{eff}}\mathbf{v})_m = \mathbf{L}_m\mathbf{v}_m + \sum_{q=-M}^M \mathbf{N}_{m-q}\mathbf{v}_q. \quad (16.2)$$

The first term requires one sparse primary-space matrix-vector product per sideband. The second term is a sideband convolution: for moderate M , direct summation; for large M , FFT-based convolution in sideband index. This is the matrix-free implementation.

16.2.4 Method 3: Block-diagonal preconditioner

The natural preconditioner for GMRES is $\mathbf{L}^{-1}$:

$$\mathbf{L}^{-1}\mathbf{A}^{\text{eff}} = \mathbf{I} + \mathbf{L}^{-1}\mathbf{N}. \quad (16.3)$$

Applying $\mathbf{L}^{-1}$ means solving $\mathbf{L}_m\mathbf{z}_m = \mathbf{v}_m$ independently for each sideband, with no coupling between sidebands. If $\|\mathbf{L}^{-1}\mathbf{N}\| < 1$, the preconditioned system is close to the identity and GMRES converges in a few iterations. Physically, this condition is met when the sideband mixing (parametric coupling) is moderate compared to the in-sideband linear response. Near gain peaks, $\mathbf{L}_m$ may be nearly singular for some m and the preconditioner degrades; in that regime a better preconditioner includes a few off-diagonal blocks.

The Schur complement $\Sigma(\Omega_m)$ inside $\mathbf{L}_m$ is applied by the nested solve of (15.8), not by explicit inversion.

16.2.5 Method 4: Frequency sweep with warm starts

The operator $\mathbf{A}^{\text{eff}}(\omega_s)$ varies smoothly with ω_s . This means the solution $\mathbf{y}(\omega_s)$ varies smoothly too. Exploiting this:

- Use the solution at frequency ω_s as the initial guess for $\omega_s + \Delta\omega$.
- For sparse direct solves, the sparsity pattern (symbolic factorization) is fixed across frequencies; only numerical re-factorization is needed.
- For Krylov methods, the previous solution initializes GMRES and may dramatically reduce iteration count.
- For very large sweeps, build a reduced basis V from a handful of representative solutions and approximate $\mathbf{y}(\omega_s) \approx V\mathbf{c}(\omega_s)$ (a rational Krylov or model-reduction approach).

16.3 Adjoint sensitivity: getting gradients for free

Suppose a device parameter p (e.g., I_c , L_0 , I_* , a coupler coefficient) appears in $\mathbf{A}^{\text{eff}}$. The transfer amplitude is $T = \boldsymbol{\ell}^\top \mathbf{y}$ where $\mathbf{A}\mathbf{y} = \mathbf{r}$. Define the adjoint variable $\mathbf{z}$ by $\mathbf{A}^\top \mathbf{z} = \boldsymbol{\ell}$. Then $T = \boldsymbol{\ell}^\top \mathbf{y} = \mathbf{z}^\top \mathbf{r}$ and the parameter gradient is:

$$\frac{\partial T}{\partial p} = -\mathbf{z}^\top \frac{\partial \mathbf{A}^{\text{eff}}}{\partial p} \mathbf{y}. \quad (16.4)$$

One forward solve and one adjoint solve yield the gradient with respect to all parameters simultaneously. This is the foundation for gradient-based fitting and for the implicit differentiation of Chapter 23: $\partial \mathbf{x}^* / \partial \boldsymbol{\theta} = -\mathbf{J}^{-1}(\partial \mathcal{R} / \partial \boldsymbol{\theta})$ applied to the gain stage.

16.4 Perturbation expansion: how gain builds up from multiple mode-conversion events

If the sideband mixing is weak ($\|\mathbf{L}^{-1}\mathbf{N}\| < 1$), expand:

$$(\mathbf{A}^{\text{eff}})^{-1} = (\mathbf{I} + \mathbf{L}^{-1}\mathbf{N})^{-1}\mathbf{L}^{-1} = [\mathbf{I} - \mathbf{L}^{-1}\mathbf{N} + (\mathbf{L}^{-1}\mathbf{N})^2 - \dots]\mathbf{L}^{-1}. \quad (16.5)$$

Therefore the transfer amplitude is:

$$T = \boldsymbol{\ell}^\top \mathbf{L}^{-1} \mathbf{r} - \boldsymbol{\ell}^\top \mathbf{L}^{-1} \mathbf{N} \mathbf{L}^{-1} \mathbf{r} + \boldsymbol{\ell}^\top (\mathbf{L}^{-1} \mathbf{N})^2 \mathbf{L}^{-1} \mathbf{r} - \dots \quad (16.6)$$

The first term is direct same-sideband propagation with no mode conversion. The second term is one conversion event (signal $\rightarrow$ idler $\rightarrow$ signal). The third term involves two events. This series explains qualitatively how gain builds up: each round trip through the conversion loop adds a power of $\mathbf{L}^{-1}\mathbf{N}$, and when $\|\mathbf{L}^{-1}\mathbf{N}\| > 1$ (above threshold), the series diverges—meaning the device amplifies. The Neumann series is not a practical numerical method for high gain, but it is an excellent interpretive tool and a starting point for perturbative analysis.

16.5 Ill-conditioning, gain peaks, and the role of residuals

The transfer amplitude $T(\omega_s) = \boldsymbol{\ell}^\top (\mathbf{A}^{\text{eff}})^{-1} \mathbf{r}$ becomes large when $\mathbf{A}^{\text{eff}}$ has a small singular value in the direction connecting $\mathbf{r}$ to $\boldsymbol{\ell}$. High gain and numerical sensitivity therefore co-locate. The forward-error heuristic

$$\frac{\|\mathbf{y} - \tilde{\mathbf{y}}\|}{\|\mathbf{y}\|} \lesssim \text{cond}(\mathbf{A}^{\text{eff}}) \frac{\|\mathbf{A}^{\text{eff}} \tilde{\mathbf{y}} - \mathbf{r}\|}{\|\mathbf{r}\|} \quad (16.7)$$

says: near gain peaks, even a small residual can produce a large error in $\mathbf{y}$, and hence in T . The practical consequence is that the residual tolerance must be tightened near gain peaks, and monitoring the condition number estimate from GMRES is worthwhile.

16.6 Common failure modes classified by their mathematical cause

Failure	Mathematical cause	Fix
No gain anywhere, all loss	Conjugate sideband missing: idler at $2\omega_p - \omega_s$ not in basis	Include $m < 0$ sidebands; use $M \geq 2$ for 4WM
Gain at wrong frequencies	Wrong sideband indexing or sign in Ω_m	Dense reference solve to verify
Finite but wrong gain	Non-converged pump state fed into gain stage	Report pump residual; gate on VALID_CONVERGED
Solver does not converge	$\mathbf{A}^{\text{eff}}$ ill-conditioned; bad preconditioner	Tighten tolerance; improve preconditioner; use direct solve
Gain jumps discontinuously	Multiple steady states; solver on wrong branch	Deflated continuation; verify pump state
Gradient computation wrong	Explicit matrix inverse used in adjoint	Use adjoint solve (16.4), not $\mathbf{A}^{-1}$
Self-energy missing	Pump-rail influence on signal rail ignored	Include $\Sigma(\Omega_m)$ via nested solves

16.7 The complete numerical algorithm

Gain at one signal frequency

1. Verify that the pump state satisfies $\|\mathcal{R}(\{\mathbf{X}_k\})\|/\|\mathbf{i}_{\text{pump}}\| < \varepsilon_p$. If not, abort.
2. For each sideband $m \in \{-M, \dots, M\}$, compute $\Omega_m = \omega_s + m\omega_p$ and assemble $\mathbf{L}_m$ via (15.5) and (15.8).
3. Assemble the Toeplitz blocks $\mathbf{N}_{m-q} = \hat{\mathbf{K}}_{m-q}^{\text{nl}}$ from the FFT of $\Gamma(t)$.
4. Build or apply $\mathbf{A}^{\text{eff}}$ using the split (16.1).
5. Solve $\mathbf{A}^{\text{eff}} \mathbf{y} = \mathbf{r}_\phi$. Record residual $\rho = \|\mathbf{A}^{\text{eff}} \mathbf{y} - \mathbf{r}_\phi\|/\|\mathbf{r}_\phi\|$.
6. Compute $T = \ell_\phi^\top \mathbf{y}$.
7. Return $G_s = 4Y_0^2 \omega_s^2 |T|^2$, ρ , and status **VALID_CONVERGED** if $\rho < \varepsilon_s$, else **FINITE_NONCONVERGED**.

16.8 The reduced textbook formula as a projection of the full operator

For completeness and to connect to the coupled-mode theory of Chapter 5, note that the familiar gain formula

$$G_s = \left| \cosh(g_e \ell) + i \frac{\kappa}{2g_e} \sinh(g_e \ell) \right|^2, \quad g_e = \sqrt{g^2 - (\kappa/2)^2} \quad (16.8)$$

is obtained from (15.9) by retaining only two sidebands (signal $m = 0$ and idler $m = 2$ for 4WM, or $m = \pm 1$ for 3WM), approximating $\mathbf{L}_m$ by a scalar phase mismatch $\kappa/2$ on the idler sideband, and treating $\mathbf{N}_{m-q}$ as a single scalar coupling constant g . In the full operator, κ is a summary of

the diagonal structure of $\mathbf{L}_m$, and g is a summary of the off-diagonal Toeplitz structure of $\mathbf{N}_{m-q}$. The reduced formula is exact only when the approximations that derive it are valid (undepleted pump, slowly-varying envelope, two participating modes, negligible harmonics). The full operator formula (15.9) requires none of these approximations.

Chapter 17

Algorithmic Recipes and Pseudocode

This chapter collects, in one place, the pseudocode for the four core algorithms and the adjoint, structured for implementation in any array language (Julia, Python/NumPy, MATLAB). The routines are written to preserve structure and avoid explicit matrix inverses.

17.1 Algorithm 1: direct uneliminated solve (reference)

Listing 17.1: Uneliminated reference gain — use only for verification on tiny systems

```
def gain_direct(omega_s, omega_p, C, G, K_lin, K_nl_hat, Y0, u_P1,
               u_P2, M):
    """
    Build and directly solve the full conversion matrix (primary +
    auxiliary blocks).
    Returns: G_s(gain), residual(relative), T(transfer amplitude)
    """
    N_sb = 2*M + 1                                # total sideband count
    Omega = [omega_s + m*omega_p for m in range(-M, M+1)]
    dim = len(u_P1)                                # internal space
    dimension

    # --- assemble full block matrix A ---
    A = zeros(N_sb*dim, N_sb*dim, complex)
    for m in range(-M, M+1):
        mi = m + M                                # row block index
        # diagonal linear block
        D_m = -Omega[mi]**2 * C + 1j*Omega[mi] * G + K_lin
        A[mi*dim:(mi+1)*dim, mi*dim:(mi+1)*dim] += D_m
        # sideband mixing (Toeplitz in m-q)
        for q in range(-M, M+1):
            qi = q + M
            ell = m - q
            if ell in K_nl_hat:
                A[mi*dim:(mi+1)*dim, qi*dim:(qi+1)*dim] += K_nl_hat[
                    ell]

    # --- right-hand side: unit signal at m=0 ---
    r = zeros(N_sb*dim, complex)
    r[M*dim:(M+1)*dim] = 2*sqrt(Y0) * u_P1        # m=0 block

    # --- solve and extract gain ---
```

```

y      = solve(A, r)                                # sparse or dense LU
T      = dot(u_P2, y[M*dim:(M+1)*dim]) * 1j*omega_s * sqrt(Y0)
G_s    = abs(T)**2
rho    = norm(A @ y - r) / norm(r)
return G_s, rho, T

```

17.2 Algorithm 2: Schur-reduced effective solve

Listing 17.2: Effective primary-space solve after Schur elimination of pump rail

```

def apply_Sigma(v, Omega, D_phi_eta, D_eta_eta, D_eta_phi):
    """Apply self-energy Sigma(Omega) to primary vector v."""
    w = D_eta_phi(Omega) @ v                # project to aux space
    z = solve(D_eta_eta(Omega), w)          # SOLVE, never invert
    return D_phi_eta(Omega) @ z             # project back

def apply_Aeff(v_sideband, omega_s, omega_p, M,
               D_phi_phi, D_phi_eta, D_eta_eta, D_eta_phi, K_nl_hat):
    """
    Matrix-free application of A_eff to a sideband-indexed vector v.
    v_sideband: list of primary vectors, indexed by m in [-M..M]
    """
    out = [zeros_like(v_sideband[0]) for _ in range(2*M+1)]
    for m in range(-M, M+1):
        mi = m + M
        Omega = omega_s + m*omega_p
        # sideband-diagonal block L_m = D_phi_phi - Sigma
        Lm_v = D_phi_phi(Omega) @ v_sideband[mi]
        Lm_v -= apply_Sigma(v_sideband[mi], Omega,
                           D_phi_eta, D_eta_eta, D_eta_phi)
        out[mi] += Lm_v
    for m in range(-M, M+1):                # Toeplitz sideband mixing
        mi = m + M
        for q in range(-M, M+1):
            qi, ell = q + M, m - q
            if ell in K_nl_hat:
                out[mi] += K_nl_hat[ell] @ v_sideband[qi]
    return out

def gain_schur(omega_s, omega_p, M, Y0, u_P1, u_P2, *operators):
    r = [zeros(...) for _ in range(2*M+1)]
    r[M] = 2*sqrt(Y0) * u_P1                # signal at m=0
    y = gmres(lambda v: apply_Aeff(v, omega_s, omega_p, M, *
                                   operators),
              r, precondition=block_diag_precond(omega_s, omega_p, M, *
                                                  operators))
    T = dot(u_P2, y[M]) * 1j*omega_s * sqrt(Y0)
    rho = norm(apply_Aeff(y,...) - r) / norm(r)
    return abs(T)**2, rho, T

```

17.3 Algorithm 3: block-diagonal preconditioner

Listing 17.3: Apply the sideband-diagonal preconditioner (inverse of L)

```

def block_diag_precond(v_sideband, omega_s, omega_p, M,
                      D_phi_phi, D_phi_eta, D_eta_eta, D_eta_phi):
    """
    Apply  $L^{-1}$ : solve  $L_m z_m = v_m$  independently for each sideband  $m$ .
    Uses Schur-nested application of  $\Sigma$ .
    """
    out = [zeros_like(v_sideband[0]) for _ in range(2*M+1)]
    for m in range(-M, M+1):
        mi = m + M
        Omega = omega_s + m*omega_p
        #  $L_m = D_{\phi\phi}(\Omega) - \Sigma(\Omega)$  is assembled as a
        # sparse matrix
        L_m = assemble_Lm(Omega, D_phi_phi, D_phi_eta, D_eta_eta,
                          D_eta_phi)
        out[mi] = solve(L_m, v_sideband[mi])
    return out

```

17.4 Algorithm 4: frequency sweep with warm starts

Listing 17.4: Signal frequency sweep exploiting smoothness

```

def gain_sweep(omega_s_list, omega_p, M, Y0, u_P1, u_P2, *operators,
               tol=1e-8):
    results = []
    y_prev = None
    for omega_s in omega_s_list:
        # assume sorted for warm start continuity
        r = build_rhs(omega_s, omega_p, M, Y0, u_P1)
        y, info = gmres(
            lambda v: apply_Aeff(v, omega_s, omega_p, M, *operators),
            r,
            x0 = y_prev,
            # warm start from previous frequency
            M = lambda v: block_diag_precond(v, omega_s, omega_p, M,
                                              *operators),
            tol = tol
        )
        T = dot(u_P2, y[M]) * 1j*omega_s * sqrt(Y0)
        rho = norm(apply_Aeff(y,...) - r) / norm(r)
        results.append({
            'omega_s': omega_s,
            'G_s': abs(T)**2,
            'rho': rho,
            'status': 'VALID_CONVERGED' if rho < tol else 'FINITE_NONCONVERGED'
        })
        y_prev = y
    return results

```

17.5 Algorithm 5: adjoint derivative for parameter fitting

Listing 17.5: Adjoint gradient of transfer amplitude w.r.t. device parameter p

```

def transfer_gradient(omega_s, omega_p, M, p, Y0, u_P1, u_P2, *
operators):
    """
    Returns dT/dp using one forward solve + one adjoint solve.
    dA_dp must return the sideband-indexed matrix d(A_eff)/d(p).
    """
    # forward solve
    r = build_rhs(omega_s, omega_p, M, Y0, u_P1)
    y = solve_Aeff(omega_s, omega_p, M, r, *operators)

    # adjoint solve:  $A^T z = ell$ 
    ell = build_ell(omega_s, M, Y0, u_P2)
    z = solve_Aeff_T(omega_s, omega_p, M, ell, *operators)

    # parameter sensitivity
    dA_dp_y = apply_dAeff_dp(y, omega_s, omega_p, M, p, *operators)
    dTdp = -dot_sideband(z, dA_dp_y)
    return dTdp

```

Chapter 18

The Space of Periodic-Steady-State Solvers

Chapters 6–12 built one solver: harmonic balance, damped Newton, continuation, condensation, conversion matrix. That is a sound default, but it is one point in a large design space, and the regime where the established Josephson-circuit tools fail to converge is exactly the regime where the less obvious members of that space were invented. The purpose of this chapter is to lay out the whole space, because the problem “find the periodic orbit of a large, strongly-forced, weakly-damped nonlinear system, then linearize about it” is not a circuit problem. It is the central computational problem of nonlinear structural dynamics, nonlinear optics, plasma equilibrium, celestial mechanics, and combustion instability, and each of those communities has built machinery the circuit community rarely cites.

Key idea

The strongly-pumped TWPA pump solve is mathematically identical to the forced periodic response of a nonlinear mechanical structure. The vibration-engineering community has spent forty years building robust continuation, stability, and bifurcation tools for precisely this object. Their tools transfer directly; the translation is purely notational.

18.1 Three independent axes

Any solver for this problem is a choice along three orthogonal axes. Mixing and matching is allowed and usually optimal.

1. **Representation of the periodic state.** *Frequency domain* (harmonic balance, Chapter 6): unknowns are Fourier coefficients; nonlinearity handled by AFT. *Time domain* (shooting, collocation): unknowns are samples or spline coefficients over one period. *Mixed* (envelope-following, multi-time): unknowns are slowly-varying envelopes of fast carriers.
2. **Globalization—how you reach the hard operating point.** None (cold Newton); line search; trust region; *continuation*—and continuation itself splits into natural-parameter, pseudo-arclength, the high-order asymptotic-numerical method, pseudo-transient, and *deflated* continuation for disconnected branches.
3. **Linear algebra inside the corrector.** Dense or sparse direct factorization; Krylov (GM-RES) matrix-free; *structure-exploiting* (block-tridiagonal along the line, Schur condensation of Chapter 9); *physics-based or learned preconditioners*.

Family	Home community	What it buys the TWPA solver
HB + AFT + Newton	RF/microwave, this book Ch. 6	The baseline; spectral accuracy in the pump.
Shooting + monodromy	dynamical systems, ODE	Time-domain PSS; stability for free; robust when HB needs many harmonics.
Orthogonal collocation (AUTO)	numerical bifurcation	Adaptive resolution of sharp transients in a period.
ANM high-order continuation	nonlinear mechanics (MAN-LAB)	Many continuation steps per factorization; sails through folds.
Hill's method	nonlinear vibration	Floquet stability from the HB solution; <i>is</i> the conversion matrix.
Deflated continuation	PDE bifurcation (Farrell)	Disconnected steady states; detects competing/parametric-oscillation branches.
Sweeping / shifted-Laplace	Helmholtz/seismic	Near-optimal inner preconditioner for the long-line solves.
Nonlinear preconditioning (ASPIN, Anderson)	domain decomposition	Tames cells with unbalanced nonlinearity (some near I_c , some not).
Neural-operator warm start	scientific ML (2025–26)	Learned initial guess for every gain-map cell; corrector still certifies.
Polynomial homotopy	numerical algebraic geometry	All steady states of the truncated model; a global safety net.

Table 18.1: The cross-domain solver landscape for the pump periodic steady state. The rows are not competitors; a production solver is a vertical stack drawn from several of them.

The remaining chapters of Part II develop the rows of Table 18.1 that are most likely to be decisive for a TWPA and least likely to be already on the reader's shelf, and close with a decision map (Section 22.2) that assembles them into one recommended stack.

Chapter 19

High-Order Continuation, Deflation, and Stability

This chapter imports the nonlinear-vibration continuation toolbox. Its three pieces—the asymptotic numerical method, deflated continuation, and Hill’s method—answer, respectively: how to walk to the operating point in very few factorizations, how to be sure you are on the right solution branch, and how to know whether the pumped state you found is even stable.

19.1 The asymptotic numerical method (ANM)

Natural and arclength continuation (Chapter 8) take *small* steps and pay one Jacobian factorization per step. The asymptotic numerical method instead represents a finite arc of the solution branch by a high-order power series and pays one factorization for the *whole arc*.

Recast the residual so every nonlinearity is at most quadratic by introducing auxiliary variables (for the Josephson branch, $u = \sin(\psi/\varphi_0)$, $w = \cos(\psi/\varphi_0)$ obey the quadratic differential relations $u' = w\psi'/\varphi_0$, $u^2 + w^2 = 1$). The continuation problem becomes $\mathcal{R}(\mathbf{U}, \lambda) = \mathbf{L}_0 + \mathbf{L}_1(\mathbf{U}) + \mathbf{Q}(\mathbf{U}, \mathbf{U}) = \mathbf{0}$ with $\mathbf{Q}$ bilinear. Expand the branch in a path parameter a ,

$$\mathbf{U}(a) = \mathbf{U}_0 + \sum_{k=1}^{N_o} a^k \mathbf{U}_k, \quad \lambda(a) = \lambda_0 + \sum_{k=1}^{N_o} a^k \lambda_k, \quad (19.1)$$

substitute, and collect powers of a . Order one gives the tangent; order k gives

$$\mathbf{J}_0 \mathbf{U}_k = (\text{known combination of lower-order terms}), \quad (19.2)$$

where $\mathbf{J}_0$ is the *same* tangent operator at every order. One factorization of $\mathbf{J}_0$ therefore yields all N_o series coefficients. The series has a finite, automatically estimable radius of convergence; the step length $a_{\max}$ is chosen so the residual stays below tolerance, and Padé resummation extends the reach further. Because the representation is a smooth series in an arclength-like parameter, ANM traverses turning points without special handling—the failure mode of natural continuation. This is the engine of the MANLAB package and is, empirically, one of the most robust path-followers available for strongly nonlinear forced systems.

Key idea

ANM amortizes the expensive operation (factorization) over a long, high-order step instead of many short first-order ones. For a pump sweep this can cut the number of factorizations along a power axis by an order of magnitude, and it crosses the folds where a gain map's sharp onset edges live.

19.2 Deflated continuation and competing branches

A strongly-pumped parametric line is not guaranteed to have a unique periodic steady state. Above threshold it can sustain a parametric oscillation; near bifurcations it can be bistable. Arclength continuation only ever sees the branch continuously connected to $\mathbf{x} = \mathbf{0}$, so a coexisting branch—and the physical instability it represents—is simply invisible, and a cold Newton solve may land on it and report a finite, plausible, wrong answer.

Deflation removes this blind spot. Given a known solution $\mathbf{x}_1^*$ of $\mathbf{R}(\mathbf{x}; \lambda) = \mathbf{0}$, replace the residual by

$$\mathcal{M}(\mathbf{x}; \mathbf{x}_1^*) \mathbf{R}(\mathbf{x}; \lambda) = \mathbf{0}, \quad \mathcal{M}(\mathbf{x}; \mathbf{x}_1^*) = \left(\frac{1}{\|\mathbf{x} - \mathbf{x}_1^*\|^p} + \sigma \right) \mathbf{I}, \quad (19.3)$$

which is singular at $\mathbf{x}_1^*$ and so forbids Newton from reconverging there; restarting from the *same* initial guess then finds a *different* root if one exists. Combined with continuation (deflated continuation) this maps disconnected bifurcation diagrams without prior knowledge of the solutions, and the deflated Newton system is solved efficiently by a Sherman–Morrison correction to the original sparse factorization. For the TWPA this is both a correctness check (am I on the operating branch the experiment sits on?) and a physics tool (where does parametric oscillation switch on?).

19.3 Hill's method: stability and the conversion matrix are the same object

Once a pump state is found, two questions remain: is it stable, and what is its small-signal gain? Both are answered by the linearization (10.1), and the nonlinear-vibration community computes them by *Hill's method*. Seek perturbations of Floquet form

$$\delta\phi(t) = e^{st} \sum_{m=-M}^M \varphi_m e^{im\omega_p t}, \quad (19.4)$$

and substitute into (10.1). Matching harmonics gives a quadratic eigenvalue problem in s whose matrices are built from the very same Fourier blocks $\hat{\mathbf{K}}_\ell$ of the pump-modulated stiffness that appear in the conversion matrix (10.3). The eigenvalues s are the Floquet exponents: $\text{Re}(s) > 0$ flags an unstable pump state (e.g. the onset of parametric self-oscillation), which is often the true reason a continuation stalls.

Key idea

Hill's matrix and the conversion matrix are assembled from identical ingredients. Setting $s = i\omega_s$ and adding a drive turns the Hill operator into the conversion matrix of Chapter 10; removing the drive and solving the homogeneous eigenproblem gives stability. Implement one assembly and get gain *and* stability from it. A solver that computes gain but not stability is leaving free, already-assembled information on the table—and that information often explains the convergence failures.

Bifurcation tracking extends this: by bordering the Hill eigenproblem with a defining condition for a fold or a Neimark–Sacker (torus) bifurcation and continuing the augmented system, one follows the *boundary* of the stable operating region directly in the (ω_p, P_p) plane—which is exactly the boundary of the usable part of a gain map.

Chapter 20

Time-Domain, Mixed, and Quasi-Periodic Routes

Harmonic balance is not the only representation, and it is not always the best one. When the pump waveform is sharp (a strongly-driven junction near I_c generates many harmonics) the Fourier truncation H must grow, and a time-domain method that resolves the sharp part adaptively can win.

20.1 Shooting and the monodromy matrix

Shooting seeks an initial condition $\mathbf{z}_0 = (\phi_0, \dot{\phi}_0)$ that reproduces itself after one pump period: $\mathbf{z}(T; \mathbf{z}_0) = \mathbf{z}_0$. Newton on this period map needs the Jacobian $\partial \mathbf{z}(T)/\partial \mathbf{z}_0$, the *monodromy matrix* $\Phi(T)$, obtained by integrating the variational equation alongside the trajectory. The Floquet multipliers are the eigenvalues of $\Phi(T)$; stability and the small-signal response follow from the same object. Thus shooting yields, by a different route, the same stability-and-conversion information as Hill's method: the monodromy matrix (time domain) and the Hill matrix (frequency domain) are two representations of the identical Floquet operator. Multiple shooting (split the period into segments solved in parallel) and orthogonal collocation (the method behind AUTO) improve robustness and allow adaptive resolution within the period; they are the tools of choice when a single global Fourier basis is inefficient.

Recipe

Use HB when the pump is moderately nonlinear and few harmonics suffice (the usual TWPA case). Switch to collocation/shooting when (i) the junction is driven so hard that H must be large, (ii) you suspect a period-doubling or torus bifurcation and want the multipliers directly, or (iii) HB continuation stalls and you want an independent check on whether a solution exists at all.

20.2 Quasi-periodic and invariant-torus methods for strong signals

The conversion matrix of Chapter 10 assumes an *undepleted* pump and an infinitesimal signal. Gain compression, intermodulation, and pump depletion appear when the signal is no longer infinitesimal, and then the state is *quasi-periodic* in two incommensurate frequencies (ω_p, ω_s) .

The right object is a two-frequency harmonic balance on the invariant torus,

$$\phi(t) = \sum_{k,\ell} \Phi_{k\ell} e^{i(k\omega_p + \ell\omega_s)t}, \quad (20.1)$$

which the ANM (Section 19.1) continues robustly in a quadratic recast without prior knowledge of the amplitudes. This is the principled route to a 1-dB-compression map and to the saturation power, quantities a single-tone conversion matrix cannot produce. It is more expensive and should be reserved for the compression study, not the linear gain map.

20.3 Envelope-following for slow modulation

When the pump itself is slowly ramped or modulated (as in a calibration sweep), envelope-following methods integrate the slow envelope of the fast-periodic state, taking steps of many pump periods. They are the mixed-representation entry of Table 18.1 and are useful for transient or pulsed-pump studies rather than steady gain maps.

Chapter 21

Structure-Exploiting and Preconditioned Linear Algebra

Inside every Newton or continuation step is a linear solve, and for a 10^4 -element line that solve dominates the cost. Chapter 9 already removed the linear nodes by Schur condensation; this chapter exploits the remaining structure and imports preconditioners from communities that solve the same operator under a different name.

21.1 The line is block-tridiagonal; the Jacobian is block-banded

A transmission line couples each cell only to its neighbours, so at each harmonic the operator $D_m = K - \omega_m^2 C + i\omega_m G$ is block-tridiagonal in the spatial index. The full HB/conversion Jacobian is block-banded in the node index and small-dense in the harmonic index (it couples harmonics differing by up to $2H$). This is the structure to respect: a block-Thomas / cyclic-reduction factorization of a block-tridiagonal system is $O(N)$ in the number of cells and parallelizes by cyclic reduction, versus the $O(N^{1.5}-N^2)$ of a generic sparse factorization. Order the unknowns node-major within harmonic-minor blocks and the factorization sees the band.

21.2 The line is a discretized wave operator: sweeping and shifted-Laplace preconditioners

Here is a transfer the circuit literature rarely makes. The diagonal block D_m is, physically, a one-dimensional Helmholtz-type operator: a wave propagating along a weakly-damped line at frequency ω_m . Long, high- Q lines make D_m indefinite and ill-conditioned, which is exactly the notorious difficulty of the Helmholtz equation in seismic imaging—and that community built near-optimal preconditioners for it. The *shifted-Laplace* preconditioner replaces D_m by a complex-shifted operator $K - (\omega_m^2 + i\epsilon)C$ that is easy to factor and clusters the preconditioned spectrum; the *sweeping* preconditioner (a moving-PML approximate factorization that marches along the propagation direction) is essentially an exact one-way factorization in 1-D and gives a near-constant GMRES iteration count independent of line length. Because the TWPA line is genuinely one-dimensional, the 1-D sweeping factorization is not an approximation to be tuned but very nearly the exact inverse, applied at $O(N)$ cost.

Key idea

Stop thinking of the inner linear solve as “a big sparse system” and start thinking of it as “a 1-D wave equation on a long line.” That reframing hands you the Helmholtz community’s sweeping and shifted-Laplace preconditioners, which are near-optimal precisely in the long, high- Q , indefinite regime where a generic preconditioner degrades.

21.3 Nonlinear preconditioning for unbalanced cells

When the pump is strong, some cells sit near I_c (highly nonlinear) while most sit far from it (nearly linear). Global Newton then takes tiny steps dictated by the few hard cells. *Nonlinear* preconditioning fixes this by solving the hard subproblems locally before the global step: additive Schwarz preconditioned inexact Newton (ASPIN) and its relatives precondition the *residual*, not just the Jacobian, so the global iteration is no longer hostage to a handful of stiff cells. Anderson acceleration and nonlinear GMRES (NGMRES) are cheaper, matrix-light alternatives that often rescue a stalling fixed-point or Picard iteration with a few lines of code. These are worth trying before reaching for a new global method.

21.4 A global safety net: polynomial homotopy

If the branch law is truncated to a polynomial (the cubic Kerr model, Chapter 3), the truncated HB residual is a polynomial system, and numerical algebraic geometry (polynomial homotopy continuation, as in HOMOTOPYCONTINUATION.JL or Bertini) can in principle enumerate *all* its solutions. This is too expensive for the full device but is a powerful diagnostic on a small fixture: it tells you how many steady states the model admits at a given pump power and whether your iterative solver is finding the physical one. Treat it as a ground-truth oracle for the multiplicity question, complementary to deflation.

Chapter 22

Learned and Hybrid Solvers, and a Decision Map

22.1 Neural-operator warm starts

The newest entry in Table 18.1 comes from scientific machine learning, and it fits the gain-map workflow unusually well. A gain map is thousands of related solves over a smooth grid in (ω_p, P_p) and parameters θ ; the converged pump state varies smoothly across that grid. Train a neural operator $\mathcal{G}_\theta : (\theta, \omega_p, P_p) \mapsto \mathbf{x}^*$ on the cells that have already converged, and use its prediction as the *initial guess* for the classical corrector on the remaining cells. This is the Nows / neural-initialized-Newton idea: the learned map supplies a high-quality warm start, the classical Newton/continuation step does the actual solving, and the convergence guarantees and the VALID_CONVERGED certificate of Chapter 12 are untouched. Reported iteration-count reductions in the PDE literature are large, and crucially the method is honest: the surrogate is never trusted on its own, only as a seed.

Key idea

The same simulator-generated pairs $(\theta, \mathbf{x}^*)$ that you must produce anyway for the SBI calibration of Chapter 25 are exactly the training data for a neural warm-start operator. The expensive simulation budget is paid once and serves two purposes: it trains the calibrator, and it trains the warm-start that makes future maps cheap. Build the data pipeline once.

Neural *preconditioners*—learning an approximate inverse to accelerate the inner GMRES—are a parallel development; they are more invasive and less mature than warm starts, and I would adopt them only after the structure-exploiting preconditioners of Chapter 21 are exhausted.

22.2 A decision map for the TWPA

Assembling the families into one recommended production stack:

1. **Assemble and condense.** Build (C, G, K, nonlinear, ports) block-wise; Schur-condense the linear nodes (Chapter 9). Order unknowns to expose the block-tridiagonal band.
2. **Walk to the operating point by continuation.** Use ANM (Section 19.1) in pump power with warm starts; fall back to pseudo-arclength near folds and to pseudo-transient if a single

cell resists. Demote Newton/Halley to the corrector.

3. **Solve the inner system by structure.** Block-tridiagonal direct factorization for moderate N ; sweeping / shifted-Laplace preconditioned GMRES (Section 21.2) for long lines; nonlinear preconditioning if cells are unbalanced.
4. **Get gain and stability together.** Assemble the Hill / conversion matrix once (Section 19.3); read off S -parameters and Floquet exponents. Use the conjugate-idler basis of Chapter 10 so gain actually appears.
5. **Check the branch.** Run deflation (Section 19.2) at a few representative cells to confirm you are on the operating branch and to locate parametric-oscillation onset; on a small fixture, use polynomial homotopy as the multiplicity oracle.
6. **Accelerate the map.** Once enough cells are converged, fit a neural-operator warm start and seed the rest of the (ω_p, P_p, θ) grid; certify every cell with the corrector.
7. **For compression studies only,** switch to two-tone / invariant-torus HB (Section 20.2).

Key idea

Ranked by payoff for this device: (1) the conjugate-idler conversion basis—without it there is no gain to compute; (2) continuation with warm starts (ANM)—without it there is no converged state to linearize; (3) network condensation and structure-exploiting inner solves—without them the map is too slow to fit; (4) Hill stability and deflation—they tell you whether the answers are physical; (5) neural warm starts—a multiplier on an already-working pipeline. Do them in that order. The inner linear-solver micro-benchmarking that tempts every project is item (3)'s smallest sub-part and should never be step one.

Part III

From Solver to Science

Chapter 23

Matching Measurements: The Inverse Problem

23.1 What is measured and what is computed

The experiment yields gain maps: measured signal gain as a function of pump frequency and pump power (and signal frequency), for one or several fabricated devices. The solver yields the same object as a function of the parameters θ (junction I_c or kinetic scale $I_\star$, C_g , C_J , coupler strengths, line lengths, loss, and disorder statistics). Calibration is the inverse map: from observed maps to θ . Two framings exist—point estimation by optimization, and posterior estimation by inference. We treat optimization here and inference in Chapter 25; the latter subsumes the former and is the right target given fabrication scatter.

23.2 Objective design

Let $\mathbf{y}^{\text{obs}}$ be the (vectorized, masked) measured gain map and $\mathbf{y}(\theta)$ the simulated one. A weighted least-squares objective is

$$\mathcal{J}(\theta) = (\mathbf{y}(\theta) - \mathbf{y}^{\text{obs}})^T \mathbf{W} (\mathbf{y}(\theta) - \mathbf{y}^{\text{obs}}) + \mathcal{P}(\theta), \quad (23.1)$$

with $\mathbf{W}$ encoding measurement precision and the masking of non-converged cells (their weight is zero), and $\mathcal{P}$ a prior/regularizer. Several design choices materially affect identifiability: match in decibels (gain is exponential, so dB linearizes the dynamic range), align maps on physical axes (internal pump power, not external, after de-embedding the line/coupler attenuation), and prefer summary features (gain-band center, peak gain, bandwidth, ripple) over raw pixels when the pixel-level model is imperfect—a point that returns in the SBI summary-statistic discussion.

23.3 Identifiability and degeneracy

Different parameter combinations can produce nearly identical gain maps; for instance, a change in C_g that shifts the dispersion can be partly compensated by a change in junction inductance. The inverse problem is therefore typically ill-conditioned, and a single best fit is misleading. The practical responses are: (i) fit several operating regimes jointly (different pump frequencies probe different parts of the dispersion), (ii) include multiple devices to separate device-to-device disorder from systematic model error, and (iii) report a posterior rather than a point, so that de-

generacies appear as correlations rather than as spurious precision. This is the central argument for SBI over plain optimization.

23.4 Differentiable simulation and the adjoint

If the solver is differentiable, gradient-based optimization of (23.1) is efficient. The forward solve has two stages, and both can be differentiated. For the pump solve, the implicit function theorem gives the parameter sensitivity of the converged state without differentiating the iteration: from $\mathbf{R}(\mathbf{x}^*(\boldsymbol{\theta}), \boldsymbol{\theta}) = \mathbf{0}$,

$$\frac{\partial \mathbf{x}^*}{\partial \boldsymbol{\theta}} = -\mathbf{J}^{-1} \frac{\partial \mathbf{R}}{\partial \boldsymbol{\theta}}, \quad (23.2)$$

reusing the already-factored Jacobian $\mathbf{J}$. The conversion-matrix solve is linear, so its sensitivity is a second linear solve. An adjoint formulation computes $\partial \mathcal{J} / \partial \boldsymbol{\theta}$ at the cost of one extra linear solve per stage regardless of the dimension of $\boldsymbol{\theta}$, which is what makes gradient-based calibration scale to many parameters. Differentiability is also exactly the property that lets the solver sit inside a neural training loop as a differentiable physics layer, should one prefer end-to-end learning to SBI.

Key idea

Do not differentiate through the Newton iteration. Differentiate the *solution* via the implicit function theorem (23.2), reusing the converged factorization. This is both cheaper and numerically cleaner, and it is available even when the iteration used line searches or continuation.

Chapter 24

Fabrication Variation and Disorder

Real lines are not uniform. Junction areas, kinetic-inductance film properties, and capacitances vary cell to cell and wafer to wafer. Disorder is not a nuisance to be averaged away; it shapes the gain ripple, the band edges, and the device-to-device spread that calibration must explain.

24.1 Modeling disorder

Represent per-cell parameters as draws from a distribution, $\theta_n = \bar{\theta} (1 + \sigma \xi_n)$ with ξ_n standardized (often correlated along the line to capture spatial gradients), and treat $(\bar{\theta}, \sigma, \text{correlation length})$ as hyperparameters. The solver already accepts per-cell parameters (the assembly of Chapter 4 carries a modulation array), so disorder is a change of inputs, not of code. Two distinct effects appear: *coherent* modulation (designed, e.g. for stopbands) and *incoherent* disorder (random), and the gain map distinguishes them—coherent modulation moves band edges predictably, disorder broadens and ripples them stochastically.

24.2 Propagating disorder to observables

Given a disorder model, the distribution of gain maps is obtained by Monte Carlo over disorder realizations. The summary statistics of that ensemble (mean gain, ripple variance, band-edge spread) are themselves the things to compare against a measured population of devices. This pushes us out of point estimation entirely: the natural calibration target is the *distribution* of maps given hyperparameters, which is precisely the likelihood-free setting SBI is built for.

Chapter 25

Calibration by Simulation-Based Inference

The forward solver is an expensive, nonlinear, stochastic (through disorder) map from parameters θ to gain maps $\mathbf{x}$. We want the posterior $p(\theta \mid \mathbf{x}^{\text{obs}})$. The likelihood $p(\mathbf{x} \mid \theta)$ is intractable—there is no closed form for the distribution of a gain map given hyperparameters—so classical likelihood-based inference does not apply. Simulation-based inference (SBI) learns the needed densities from simulations alone.

25.1 The SBI workflow

SBI replaces the likelihood with a neural density estimator trained on simulated pairs. The canonical workflow is:

1. **Prior.** Specify $p(\theta)$ over physical parameters and disorder hyperparameters, informed by fabrication knowledge and the materials priors of Chapter 24.
2. **Simulate.** Draw $\theta^{(j)} \sim p(\theta)$ and run the solver to produce $\mathbf{x}^{(j)}$ (optionally reduced to physically meaningful summary statistics $\mathbf{s}^{(j)} = S(\mathbf{x}^{(j)})$), building a training set $\{(\theta^{(j)}, \mathbf{x}^{(j)})\}$.
3. **Train.** Fit a neural density estimator (typically a normalizing flow) to approximate the posterior (neural posterior estimation, NPE), the likelihood (NLE), or the likelihood-to-evidence ratio (NRE).
4. **Infer.** Evaluate the trained estimator at the observed map $\mathbf{x}^{\text{obs}}$ to obtain $p(\theta \mid \mathbf{x}^{\text{obs}})$; for NLE/NRE a sampling step (MCMC) constructs the posterior, while NPE yields it directly.

The methods are likelihood-free: they need only the ability to simulate, not gradients through the simulator, so they are compatible with the black-box solver even before it is made differentiable.

25.2 Amortization: the payoff for an expensive solver

The decisive property for this application is *amortization*. Amortized NPE trains once on a broad set of simulations and then returns a posterior for *any* new observation almost instantaneously, with near-fixed cost per evaluation. Concretely: pay the simulation cost once to build the training set, train the flow, and thereafter calibrate each newly measured device in well under a second,

including full uncertainty. Compared with running an expensive optimization (or MCMC that calls the solver thousands of times) per device, amortized SBI moves the cost off the critical path and makes calibrating a *population* of devices feasible. Sequential variants (SNPE/SNLE/SNRE) instead focus all simulations on one observation for maximal simulation efficiency when only a single device matters; the choice between amortized and sequential is a budget decision.

Key idea

SBI is the natural fit because (i) the likelihood of a gain map given fabrication hyperparameters is intractable but the map is simulable; (ii) disorder makes the problem inherently distributional, which SBI handles by construction; (iii) amortization turns the expensive solver into a one-time training cost and gives near-instant, uncertainty-quantified calibration for every subsequent device—exactly the regime of “calibrate many devices, then predict a more complex one.”

25.3 Summary statistics and the role of physics

Feeding raw gain maps (thousands of pixels, many non-converged) to a density estimator is wasteful and fragile. Physically motivated summary statistics—peak gain, gain-band center and width, ripple amplitude and period, band-edge frequencies, idler-conversion ratio—compress the observation to a low-dimensional, interpretable vector and let the estimator focus on what carries information about θ . Choosing summaries is where the physics of Parts I–II pays off in the inference: a summary that a coupled-mode argument shows to depend monotonically on β_{nl} or on the phase-matching detuning is far more identifying than a raw pixel.

25.4 Calibrating the calibrator

An SBI posterior must itself be checked for calibration: does a stated 90% credible region contain the truth 90% of the time? Standard diagnostics—expected coverage, posterior shrinkage, and posterior z-scores on held-out simulations—quantify whether the posterior is well-calibrated, over-confident, or conservative. Reporting these alongside the inferred device parameters is what makes the calibration scientifically defensible rather than merely a fit with error bars of unknown meaning.

25.5 From one device to a more complex TWPA

The endpoint of the program is to use parameters and fabrication statistics inferred on simpler devices to predict, with quantified uncertainty, the behavior of a more complex amplifier. SBI supports this directly: the inferred posterior over physical and disorder hyperparameters becomes the prior for the complex device, the solver propagates that prior through the new topology, and the predictive distribution of the complex device’s gain map follows by forward simulation. Disorder hyperparameters transfer across designs (they are properties of the fabrication process, not the layout), which is precisely what makes inference on simple devices informative about complex ones.

Chapter 26

Software Architecture and Roadmap

26.1 The block-residual interface

The topology-invariance of (2.5) licenses a clean architecture in which every block—port, coupler, JTL section, kinetic-inductance section—implements the same interface: contribute to $\mathbf{C}$, $\mathbf{G}$, $\mathbf{K}$; evaluate the local nonlinear current and its tangent; and supply source terms. The global residual is the assembled sum, and the pump solver, the conversion-matrix stage, and any time-domain validator all consume the same assembled objects. This separation is what replaces dependence on an opaque backend with a transparent, testable solver, and it is what makes the kinetic-inductance extension a new block rather than a new program.

Recipe

Block interface (per block b): `assemble_C`, `assemble_G`, `assemble_K`, `current(phi)` returning $\mathbf{B}_b \mathbf{i}_b(\mathbf{B}_b^\top \phi)$, `tangent(phi)` returning the instantaneous stiffness, and `source(t)`. Assemble once into $(\mathbf{C}, \mathbf{G}, \mathbf{K}, \text{nonlinear map, ports})$; hand those to the pump solver and the conversion stage.

26.2 Language and differentiation

The bottleneck is algorithmic (the monolithic nonlinear system), not the language. The arguments for staying in a high-performance array language with mature sparse linear algebra and a trusted reference solver are strong; the arguments for a differentiable array framework appear once gradient-based calibration or end-to-end learning becomes the priority. A pragmatic split is to keep the trusted reference and the production pump/conversion solver in the performance language, and to expose the forward map (or a reduced surrogate) to the inference layer through either implicit differentiation (23.2) or a black-box SBI interface that needs no gradients at all. Either way, the network condensation of Chapter 9 and the continuation of Chapter 8 are the components that determine whether the whole pipeline is feasible.

26.3 A concrete build order

1. Assemble $(\mathbf{C}, \mathbf{G}, \mathbf{K}, \text{nonlinear, ports})$ from blocks; verify the linear limit against ABCD (Chapter 12, step 1).

2. Implement AFT residual and analytic Jacobian (Chapter 6); verify the Jacobian by finite difference on a small fixture.
3. Build the conversion matrix with the conjugate idler and $M \geq 2$ (Chapter 10); verify zero/weak-pump limits and reproduce coupled-mode gain (Chapter 12, steps 2–4). *This is the step that first produces gain.*
4. Add pump-power continuation with warm starts (Chapter 8); confirm the operating regime is now reachable.
5. Add network condensation (Chapter 9); confirm scaling to 10^4 elements.
6. Establish the cross-tool agreement region, then push beyond it (Chapter 12, step 6).
7. Wrap the verified forward map for calibration: objective and implicit-diff gradients (Chapter 23), then amortized SBI over physical and disorder hyperparameters (Chapters 24–25).
8. Add the kinetic-inductance block (Chapter 3) as a parameter/branch-law change; re-run the validation ladder.

Key idea

Order matters. The most common failure of TWPA-solver projects is to invest in pump-solver variety before the conversion matrix produces gain, and in benchmarking inner linear solvers before continuation and condensation are in place. Build gain first (step 3), make it converge second (step 4), make it scale third (step 5), then earn trust (step 6) before any calibration.

Appendix A

Notation and Conventions

Symbol	Meaning
$\phi, \mathbf{v} = \dot{\phi}$	node-flux vector; node-voltage vector
$\boldsymbol{\eta}$	auxiliary pump-rail node fluxes
$\mathbf{x} = (\phi, \boldsymbol{\eta})^\top$	full internal state vector
$\Phi, H, X = \Phi \oplus H$	primary space, auxiliary space, internal direct-sum space
$\mathbf{B}, \mathbf{D}, \mathbf{B}_J, \mathbf{B}_\phi$	incidence matrices (all, series, Josephson, nonlinear)
$\mathbf{C}, \mathbf{G}, \mathbf{K}_{\text{lin}}$	global capacitance, conductance, linear stiffness
$\mathbf{C}_{ab}, \mathbf{G}_{ab}, \mathbf{K}_{ab}$	block components of above in $\Phi \oplus H$ split
$\mathbf{D}_{ab}(\Omega) = -\Omega^2 \mathbf{C}_{ab} + i\Omega \mathbf{G}_{ab} + \mathbf{K}_{ab}^{\text{lin}}$	sideband-diagonal matrix polynomial
ψ_e	branch flux of branch e ; $\boldsymbol{\psi} = \mathbf{B}^\top \phi$
$\varphi_0 = \Phi_0/2\pi$	reduced flux quantum; $\Phi_0 = h/2e$
$I_c, L_{J0} = \varphi_0/I_c$	junction critical current; zero-bias Josephson inductance
I_*, I'_*, α_K	KI scale current; effective KI parameter; kinetic fraction
β_{nl}	nonlinearity shape factor
$\Gamma(t), \Gamma_n(t)$	time-periodic differential inverse inductance (full; per branch)
$\omega_p, \omega_s, \omega_i$	pump, signal, idler angular frequencies
$T = 2\pi/\omega_p$	pump period
H, M	pump-harmonic truncation; conversion-sideband truncation
$m, q \in \mathcal{M} = \{-M, \dots, M\}$	sideband indices
$\Omega_m = \omega_s + m\omega_p$	sideband frequency
$\hat{\mathbf{K}}_\ell$	ℓ -th Fourier coefficient of time-periodic stiffness $\mathbf{K}(t)$
$\hat{\mathbf{K}}_{m-q}^{\text{nl}}$	nonlinear part of $\hat{\mathbf{K}}_{m-q}$; Toeplitz sideband mixer
$\mathbf{A}_{mq}$	full conversion matrix block
$\mathbf{A}_{mq}^{\text{eff}} = \mathbf{L}_m \delta_{mq} + \mathbf{N}_{m-q}$	Schur-reduced effective primary-space operator
$\Sigma(\Omega)$	self-energy: Schur complement loading from aux. rail elimination
$\mathbf{L}_m = \mathbf{D}_{\phi\phi}(\Omega_m) - \Sigma(\Omega_m)$	effective primary diagonal block
$\mathbf{N}_{m-q} = \hat{\mathbf{K}}_{m-q}^{\text{nl}}$	sideband-mixing Toeplitz block
E_0	embedding operator into sideband $m = 0$ block
$\mathbf{u}_{P_1}, \mathbf{u}_{P_2}$	input and output port selectors
$\mathbf{r} = E_0 \mathbf{u}_{P_1}$	mathematical input vector
$\boldsymbol{\ell}^\top = \mathbf{u}_{P_2}^\top E_0^\top$	mathematical output functional
$T(\omega_s) = \boldsymbol{\ell}^\top (\mathbf{A}^{\text{eff}})^{-1} \mathbf{r}$	transfer amplitude
$G_s = 4Y_0^2 \omega_s^2 T ^2$	signal power gain
$Y_0 = 1/Z_0$	reference admittance
a_p, b_p	incident/outgoing power waves at port p
$\Delta k, \kappa, g_e$	phase mismatch; total mismatch; effective gain coefficient (CME)

Symbol	Meaning
$\boldsymbol{\theta}$	physical and disorder parameters to be calibrated
$p(\boldsymbol{\theta} \boldsymbol{x}^{\text{obs}})$	posterior distribution from SBI

Appendix B

ABCD, Impedance, and Scattering Conversions

For a two-port with ABCD matrix $\mathbf{T} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ and equal real reference impedance Z_0 ,

$$\mathbf{S} = \frac{1}{A + B/Z_0 + CZ_0 + D} \begin{bmatrix} A + B/Z_0 - CZ_0 - D & 2(AD - BC) \\ 2 & -A + B/Z_0 - CZ_0 + D \end{bmatrix}. \quad (\text{B.1})$$

A series impedance Z has $\mathbf{T} = \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$; a shunt admittance Y has $\mathbf{T} = \begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix}$. A cascade is the ordered product $\mathbf{T} = \prod_n \mathbf{T}_n$. For a multiport with admittance matrix $\mathbf{Y}(\omega)$ and common reference Z_0 ,

$$\mathbf{S} = (\mathbf{I} - Z_0 \mathbf{Y})(\mathbf{I} + Z_0 \mathbf{Y})^{-1}. \quad (\text{B.2})$$

With a diagonal matrix of unequal port impedances $\mathbf{Z}_0$, $\mathbf{S} = (\mathbf{I} - \mathbf{Z}_0^{1/2} \mathbf{Y} \mathbf{Z}_0^{1/2})(\mathbf{I} + \mathbf{Z}_0^{1/2} \mathbf{Y} \mathbf{Z}_0^{1/2})^{-1}$.

The Schur-complement self-energy $\Sigma(\Omega)$ of Chapter 15 is related to the two-port S-parameter of the auxiliary network seen from the coupler ports by the Y-to-S conversion above, with Z_0 the reference impedance at each coupler port.

Appendix C

Units, Constants, and the Reduced Flux Quantum

The flux quantum is $\Phi_0 = h/2e \approx 2.0678 \times 10^{-15}$ Wb and the reduced flux quantum is $\varphi_0 = \Phi_0/2\pi \approx 3.2911 \times 10^{-16}$ Wb. The Josephson inductance is $L_{J0} = \varphi_0/I_c$; for $L_{J0} = 79$ pH, $I_c \approx 4.16 \mu\text{A}$. Pump power in dBm maps to peak current into Z_0 by $P[\text{W}] = 10^{-3} \cdot 10^{P[\text{dBm}]/10}$, $I_{\text{peak}} = \sqrt{2P/Z_0}$.

For kinetic inductance, $L_0 = L_G + L_{K0}$, $\alpha_K = L_{K0}/L_0$ is the kinetic fraction, and the effective nonlinear scale is $I'_* = I_*/\sqrt{\alpha_K}$. The Mena et al. (2026) experimental result establishes that $I'_* \sim I_{\text{dep}}$ (the depairing current), not I_c , in high-quality NbTiN films. The depairing current from Ginzburg–Landau theory for a strip of cross-section A is $I_{\text{dep}} = J_{\text{dep}} \cdot A$ with $J_{\text{dep}} = \Phi_0/(3\sqrt{3}\pi\mu_0\lambda^2\xi)$.

Appendix D

Useful Fourier and AFT Identities

For a real T -periodic signal sampled at $t_m = mT/N_t$, the cosine/sine coefficients follow the AFT projection (6.4); reconstruction is $f(t_m) = \frac{1}{2}a_0 + \sum_{k \geq 1} [a_k \cos(k\omega_p t_m) + b_k \sin(k\omega_p t_m)]$. To represent a nonlinearity generating up to harmonic $H_{\max}$ without aliasing, require $N_t \geq 2H_{\max} + 1$; oversampling ($N_t \gtrsim 4(2H + 1)$) reduces projection error.

The differential stiffness $\Gamma(t)$ for a Josephson rail (evaluated on the pump state) has Fourier components up to harmonic H (matching the pump HB truncation), so the Toeplitz sideband-mixing blocks $\mathbf{N}_{m-q} = \hat{\mathbf{K}}_{m-q}^{\text{nl}}$ are nonzero for $|m - q| \leq 2H$. The sideband truncation M should satisfy $M \geq 2H$ so that all nonzero $\mathbf{N}_{m-q}$ fit inside the sideband lattice. For a kinetic-inductance rail (polynomial nonlinearity), the range of nonzero Fourier components of $\Gamma(t)$ is finite and set by the truncation order of the pump solve.

The idler sideband for 4WM: signal at ω_s , pump at ω_p , idler at $\omega_i = 2\omega_p - \omega_s = \omega_s + 2\omega_p - 2\omega_s$. In the sideband lattice $\Omega_m = \omega_s + m\omega_p$, the idler lives at $m = 2$ on the *conjugate* branch (negative-frequency component). For the sideband basis to include the idler, one needs both positive and negative m , and $M \geq 2$.

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